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Key Points:

- Within NASA's Artemis candidate landing regions, we identified 130 candidate landing sites that satisfy the Human Landing System's requirements
- Illumination modeling revealed significant temporal and spatial variability, indicating that depending on the launch year, different landing sites will be available
- Volatile estimates of accessible permanently shadowed regions (PSRs) vary greatly and are largely controlled by surface temperature and PSR age

Supporting Information:

Supporting Information may be found in the online version of this article.

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Assessing Potential Landing Sites With Favorable Illumination and Accessible, Potentially Volatile-Rich Permanently Shadowed Regions Within Artemis Candidate Landing Regions

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Abstract The lunar south pole features permanently shadowed regions (PSRs) that never receive direct sunlight. These regions maintain cold temperatures that potentially trap and accumulate volatiles over geologic timescales. Within NASA's Artemis candidate landing regions, we identified 130 candidate landing sites (CLSs) that satisfy the Human Landing System's requirements and provide access to PSRs located within 2 km, the current radial limit for astronauts without a rover. We modeled illumination conditions for these sites during the midsummers from 2025 to 2032 to ensure they remain sunlit throughout 6.5 days, the nominal duration of the first Artemis surface missions. The modeling revealed significant temporal and spatial variability in illumination among the Artemis landing regions. Thus, depending on the launch year, different landing sites will be available. Additionally, we approximated the volatile content in accessible PSRs by combining Diviner-derived temperature data with model ages of PSR-hosting craters estimated using a topographic diffusion model. The model suggests 85% of PSR-hosting craters near CLSs are younger than 1,500 Ma. Cumulative volatile estimates vary greatly among different CLSs and are largely controlled by surface temperature and PSR age; colder and/or older PSRs have the greatest volatile trapping and preservation potential. Although some landing sites have PSRs potentially containing significant amounts of water, key factors such as volatile depth remain uncertain, making their in situ resource utilization potential difficult to assess. Our study will provide critical input for Artemis landing site selection and preparation for future missions that utilize the Moon's resources for scientific and exploratory purposes.

Plain Language Summary The south pole of the Moon is home to permanently shadowed regions (PSRs), which never see direct sunlight and can remain extremely cold. PSRs near sites traversable by astronauts can be as large as football stadiums on Earth. These cold regions may have trapped volatiles, such as water ice, for billions of years. Identifying optimal landing sites near PSRs will support future astronauts, both in studying the Moon's history and in utilizing its resources. For NASA's Artemis program, we identified 130 candidate landing sites by searching for safe, flat regions that receive sunlight for at least one week of operations and PSRs within a walking distance of 2 km. Our results show that only a few landing sites have PSRs that are both very cold and very old—prime spots for preserving ancient water ice. We approximated the amount of volatiles in each PSR, suggesting that some landing sites could provide astronauts with access to significant water ice resources. However, much remains unknown about the form and storage depth of lunar water ice. Our findings will help mission architects to select landing sites where astronauts can safely explore, collect unique samples, and prepare for future missions that could utilize the Moon's resources.

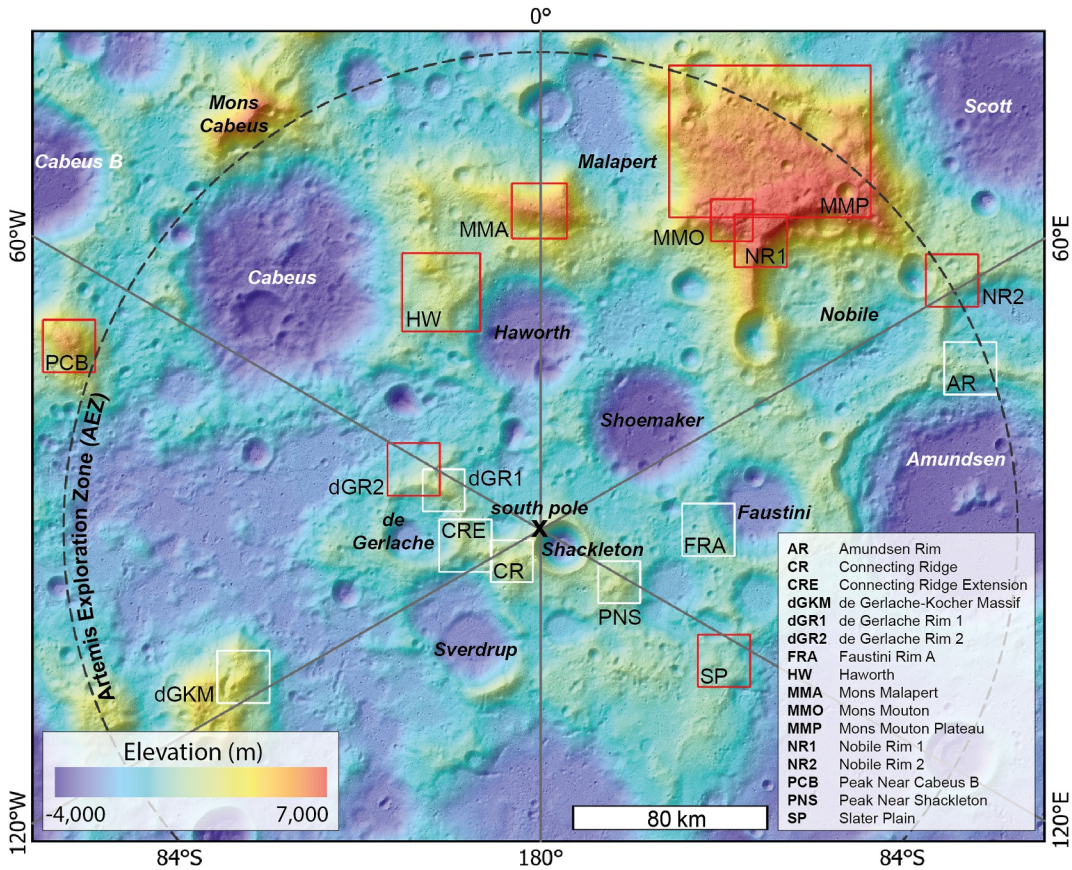


Figure 1. Geographic setting of the lunar south polar region with the nine current Artemis III candidate landing regions (CLRs) (red boxes; NASA, 2024) and previously announced CLRs (white boxes; NASA, 2022a). The base map is the 20 m/pixel Lunar Reconnaissance Orbiter's LDEM overlain on a hillshade map (Barker et al., 2023).

1. Introduction

After more than five decades of human presence confined to low-Earth orbit, several space agencies (e.g., from the United States, Europe, India, or China) have initiated exploratory endeavors to the Moon (e.g., Artemis III Science Definition Team, 2020; Carpenter et al., 2012; Hu et al., 2023; Kanu et al., 2024). While the Apollo human exploration program targeted the nearside equatorial region of the Moon, current human exploration programs are focusing on south polar destinations. The enormous scientific interest in the south polar region is linked to its unexplored terrain, its proximity to the ancient South Pole-Aitken (SPA) basin, potential volatile resources in permanently shadowed regions (PSRs), and the presence of regions that receive sunlight for most of the lunar year (e.g., NASA, 2022b).

The NASA-led Artemis program aims to land a new generation of lunar astronauts on the Moon with the Artemis III mission (NASA, 2020a, 2022a, 2024). In August 2022, NASA released a list of 13 candidate landing regions (CLRs; NASA, 2022a) within the initial Artemis exploration zone (AEZ; >84°S) that were narrowed down to nine CLRs two years later (NASA, 2024; Figure 1). This study focuses on the identification of candidate landing sites (CLSs) within these regions that fulfill the landing requirements for the Human Landing System (HLS), such as having surface slopes of less than 8° (NASA, 2019). Additionally, we assess the temporal variation in illumination conditions at these landing sites that limits their availability depending on the launch dates of the Artemis missions. For each CLS, we assess how the simultaneous visibility of the Sun and Earth constrains the availability of landing sites for future exploration throughout the midsummer periods from 2025 to 2032, a reasonable period for the Artemis landings (NASA, 2020b). The coldest PSRs are considered to have the highest potential for trapping and preserving volatile compounds, including water ice, due to their extremely low and stable temperatures (e.g., Vasavada et al., 1999; Watson et al., 1961; Williams et al., 2019). In contrast, the oldest

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PSRs may have accumulated volatiles over extended geologic timescales, potentially preserving a longer and more complete record of volatile delivery and retention on the Moon (e.g., Arnold, 1979; M. A. Siegler et al., 2011; M. Siegler et al., 2015; Watson et al., 1961). Therefore, we identified CLSs that may provide access to the coldest and oldest PSRs near the lunar poles—each located within 2 km of the landing site, the maximum range an Artemis III astronaut can explore (NASA, 2019)—by integrating temperature data with a newly developed crater diffusion model (Fassett et al., 2018, 2022; Fassett & Thomson, 2014). Accessing these sites is essential for advancing our understanding of the origin, distribution, and long-term stability of lunar polar volatiles, a key science objective outlined in the Scientific Context for the Exploration of the Moon (National Research Council, 2007) and recommended in the Decadal Strategy for Planetary Science and Astrobiology (National Academies of Sciences, Engineering, and Medicine, 2023).

2. Data and Methodology

2.1. Data sets

To initially assess the Artemis CLRs, we used image data from the Lunar Reconnaissance Orbiter's (LRO) Wide Angle Camera with a spatial resolution of 100 m/pixel (Robinson et al., 2010), and the Narrow Angle Camera (NAC) south polar mosaics, which have a resolution of ∼1 m/pixel (Archinal et al., 2023; Wagner et al., 2015). We further incorporated LRO Lunar Orbiter Laser Altimeter (LOLA) digital elevation models (LDEM) at a resolution of 20 m/pixel (Barker et al., 2023; Mazarico et al., 2011; Smith et al., 2010) to provide topographic context. We utilized these DEMs to (a) generate artificially illuminated hillshade maps at varying azimuths to minimize potential biases caused by the challenging and variable illumination conditions in the south polar region and (b) identify the location and extent of PSRs (Barker et al., 2023). Furthermore, we used LOLA Reduced Data Record data (LOLA RDR; e.g., Mazarico et al., 2012; Gläser et al., 2014) for diffusion modeling to ensure that topographically accurate crater shapes were used as model input. LOLA RDR tracks are available in the Planetary Data System (Lunar Orbital Data Explorer–Tools) and provide individual altimetry readings that sample ∼5 m spots with ∼10 cm precision along the spacecraft's orbit track (Smith et al., 2010). South polar temperature maps with a spatial resolution of 240 m/pixel (Williams et al., 2019) were obtained from the Diviner Lunar Radiometer Experiment, which measures the Moon's infrared emissions (Paige, Foote, et al., 2010; Williams et al., 2019). We used the abovementioned data sets to study the thermal conditions within PSRs, identify the coldest ones within an accessible distance of potential landing sites, and ultimately assess their potential for preserving volatiles.

2.2. Identification of Candidate Landing Sites

To identify potential landing sites within the Artemis CLRs, technical requirements for landing such as slope and illumination must be taken into account. The HLS has been tentatively selected to carry the Artemis III astronauts to the lunar surface. As defined for the Artemis III HLS (NASA, 2019), the landing site slope "shall provide a vertical orientation of 0–8° (threshold) and 0–5° (goal)." Given the HLS landing accuracy of 100 m (3σ) of a target landing site (NASA, 2019), the selected landing site should be at least 100 m away from slopes steeper than 8°. If the HLS is capable of adjusting its vertical tilt, it may tolerate steeper surface slopes, which would increase the number of viable landing sites. However, since this capability is unknown, our analysis is limited to slopes between 0 and 8°, following Kring et al. (2023).

We generated slope maps for each of the CLRs from the 20 m/pixel resolution LDEM (Barker et al., 2023) and rejected all areas that have surfaces steeper than 8°. Considering the Artemis III HLS landing accuracy of 100 m (NASA, 2019), we buffered the 0–8° slope areas by −100 m, identifying suitable areas for landing that meet the HLS requirements of having gentle slopes (less than 8°) and being at least 100 m away from steeper slopes. Based on this, we manually selected representative CLSs that all have shallow slopes (<5°) and, if present, are in or near highly illuminated areas. The number and location of the representative landing sites chosen for each Artemis CLR are intended to reflect the spatial variability of suitable landing areas. As illumination conditions are an additional critical constraint for landing and exploration activities, we modeled their temporal variation in the next step (Section 2.3).

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2.3. Illumination Modeling

The low angle of solar incidence and the topography of the south polar region control the illumination conditions in the AEZ in a much more complex pattern than the Apollo astronauts experienced in the equatorial region. Therefore, illumination conditions play a critical role in the selection of the landing site, as they can affect the landing approach and surface operations (e.g., Eppler, 1991; Iqbal et al., 2024; E. M. Jones, 1995; Kring et al., 2025). We investigated the illumination conditions for a specific time period rather than using the averaged Sun visibility for a lunar year (e.g., Mazarico et al., 2011). Because the Moon's rotational axis is tilted by 1.59° with respect to the ecliptic plane (Vasavada et al., 2012), it experiences seasons that significantly affect the illumination conditions near the poles. If the point where the Sun is perpendicular to the surface, referred to as the subsolar latitude, is in the southern hemisphere, it is defined as southern summer, and vice versa. Since the best illumination conditions for the south polar region are during southern summer, we calculated when midsummer (subsolar latitude $\leq -1.5°$; Mahanti et al., 2024) occurs in the southern hemisphere from 2025 to 2032 using JPL Planetary and Lunar Ephemeris data "DE441" (Park et al., 2021).

Since the Artemis III crew will be on the lunar surface for 6.5 days (NASA, 2019), it may be preferable to maintain illumination throughout the duration of the mission. Therefore, we modeled the illumination conditions utilizing the "Lunar Quickmap TerrainShadows" software developed by the Applied Coherent Technology Corporation (ACT) in 2D cartographic mode with 60 m/pixel spatial resolution and 12-hr steps. We considered a surface height offset of two m, as suggested for the first generation of vertically deployed solar panels near the south pole (Ross et al., 2023), to assess solar power availability for surface missions. Due to the combination of low subsolar latitude (i.e., the Sun is visibly very close to the horizon) and topographically complex terrain in the south polar region, with elevation changes up to 10 km, a shadow model with a 150 km maximum distance was employed to account for the light rays that get intercepted by high topography far away from the CLS.

Being in sight of Earth can enable direct-to-Earth (DTE) communication and may be an important constraint for early south polar landing missions without an orbiting relay satellite. Therefore, we calculated (a) solar illumination and (b) simultaneous Sun and Earth visibility for each Artemis CLR and each midsummer period from 2025 to 2032. To extract the maximum number of days with consecutive illumination, we identified consecutive sunlit days (non-zero values) for each pixel of the illumination conditions in 12-hr steps generated before. For each pixel, our output contains the maximum number of days with consecutive (a) Sun visibility and (b) combined Sun and Earth visibility. Finally, we retrieved the number of consecutive illuminated days for all CLSs for the midsummer periods from 2025 until 2032 (see Section 4).

2.4. Diffusion Modeling

Sampling and analyzing lunar volatiles in PSRs are key scientific objectives for future south polar landing missions and a major step toward in situ resource utilization (ISRU; National Research Council, 2007). Most of the identified present-day PSRs in the lunar polar regions are within small impact craters (Bussey et al., 1999, 2010; Kumari et al., 2022), and under constant solar inclination, it can be assumed that the age of the host crater is the maximum age of the PSR within it. Because older PSRs were potentially able to accumulate larger amounts of volatiles than younger ones, determining the age of the host crater, and thus, the age of the PSR is critical for identifying the oldest PSRs that lie within the ∼2 km exploration zone around each CLS.

We used a topographic diffusion model to determine the age of the PSR-hosting craters, as other techniques, such as performing crater size-frequency distribution measurements, cannot confidently be used to derive ages for impact craters <1 km in diameter that are prevalent across the Artemis CLRs (Neukum et al., 2001; van der Bogert et al., 2025). However, the diffusion model can only be applied to small craters that were not extensively modified by erosion or superposition (Du et al., 2019; Fassett & Thomson, 2014). For this reason, the craters of interest were carefully chosen to avoid those significantly modified by overlapping impacts and ejecta infilling. Pervasive erosion is caused by a combination of persistent impact-derived erosion of hillslopes, mass wasting triggered by impact-induced seismicity, and thermal contraction-relaxation cycles operating within the uppermost lunar surface (e.g., Craddock & Howard, 2000; Fassett & Thomson, 2014; Molaro & Byrne, 2012; Richardson et al., 2005; Schultz & Gault, 1975; Soderblom, 1970). The net effect of these processes is the mobilization and transport of regolith along elevation gradients, eroding regions of higher elevation (such as crater rim crests) and

filling up low-lying areas (crater floors), resulting in the slow attenuation of topographic amplitude. Following Fick's law (e.g., Culling, 1960; Pelletier, 2008), this erosion can be modeled as topographic diffusion (Equation 1), and with the surface topography h and the diffusivity κ being known, the change in crater shape as a function of time can be determined:

$$\frac{\partial h}{\partial t} = \kappa \nabla^2 h = \kappa \frac{\partial^2 h}{\partial x^2}. \quad (1)$$

Following Fassett et al. (2022), we consider the time taken for a crater to reach its current degraded shape as the age of the crater. Diffusivity varies as a function of crater diameter, with larger craters (km-sized) taking longer to reach degraded shapes compared to their smaller counterparts (Bouley & Baratoux, 2011; Fassett et al., 2018; Fassett & Thomson, 2014). This scale dependency is incorporated using Soderblom's (1970) power-law (Equation 2), which describes the flux of eroding craters (denoted by the constants α and β) as a function of crater size ($D_{\max}$ is the largest diameter of the craters that erode the parent crater):

$$\kappa = \frac{0.5\alpha\beta\left(\frac{D_{\max}}{2}\right)^{4-\beta}}{4-\beta}. \quad (2)$$

To determine the erosion-causing crater flux, we use the production function derived by Fassett et al. (2022) by merging the Grün et al. (1985) micrometeoroid flux with the Neukum et al. (2001) production function (see Table S1 in Supporting Information S1). The model is valid for eroding crater sizes between 1 cm and 300 m, translating to a realistically simulated erosion of PSR-hosting craters up to ~3 km in diameter (eroding crater size ≤1/10 of the eroded crater size). We set the initial crater shape following the method of Fassett and Thomson (2014) and used the depth-to-diameter ratio (d/D) of fresh craters from Stopar et al. (2017):

$$\left(\frac{d}{D}\right)_{\text{fresh}} = \begin{cases} 0.11\ ;\ 10 < D_0 \leq 40 \\ 0.13\ ;\ 40 < D_0 \leq 100 \\ 0.15\ ;\ 100 < D_0 \leq 200 \\ 0.17\ ;\ 200 < D_0 \leq 400 \\ 0.21\ ;\ D_0 > 400 \end{cases} \quad (3)$$

The diffusion of craters with a range of initial diameters was modeled at a fixed time-step over a timeframe of 3.6 Ga or until the diffused crater d/D falls below 0.04 (e.g., Basilevsky et al., 2014; Fassett et al., 2022; Mahanti et al., 2018; Minton et al., 2019).

To accurately measure the shape of the PSR-hosting craters, we used LOLA RDR tracks (e.g., Gläser et al., 2014; Mazarico et al., 2012), which provide individual altimetry readings along the spacecraft's orbit track without interpolating the data (i.e., for generating DEMs). We then applied local slope corrections to the acquired elevation profiles, because the diffusion model assumes craters formed on flat surfaces, rather than the complex background slopes in the south polar region. We considered three types of local slope corrections: (a) craters on topographic ridges (hilltops) were detrended using quadratic fitting; (b) craters on slopes were detrended using linear fitting; and (c) craters on nearly flat surfaces were normalized by mean crater elevation. Next, we compared the preprocessed crater shapes derived from the LOLA tracks to the modeled crater shapes to find the modeled degradation state that best approximates the current topographic state of the PSR-hosting crater. The time the modeled crater takes to erode from the initial shape to the current shape is considered to be the age of the crater, and thus, an upper bound age of the PSR hosted in it (see Section 6.1).

More detailed descriptions of parameters, the local slope correction, and the numerical implementation are provided in Supporting Information S1.

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Table 1
For Each of the Artemis Candidate Landing Regions, the Area Size, Fractional Area, and the Number of Representative Landing Sites That Meet the HLS Requirement of Slopes Shallower Than 8° and a Distance of More Than 100 m From Steeper Slopes Are Presented

Artemis candidate landing region (NASA, 2022a, 2024)	Area suitable for landing	Fractional area suitable for landing	No. of candidate landing sites meeting HLS requirements
Amundsen Rim (AR)	24.2 km^2	6.0%	10
Connecting Ridge (CR)	11.8 km^2	4.6%	9
Connecting Ridge Extension (CRE)	7.5 km^2	1.9%	6
de Gerlache-Kocher Massif (dGKM)	16.0 km^2	4.0%	7
de Gerlache Rim 1 (dGR1)	6.4 km^2	2.5%	6
de Gerlache Rim 2 (dGR2)	30.1 km^2	7.5%	7
Faustini Rim A (FRA)	51.8 km^2	12.9%	8
Haworth (HW)	77.3 km^2	8.7%	11
Mons Malapert (MMA)	7.1 km^2	1.6%	5
Mons Mouton (MMO)	20.7 km^2	8.1%	10
Mons Mouton Plateau (MMP)	564.8 km^2	12.7%	11
Nobile Rim 1 (NR1)	22.2 km^2	5.5%	9
Nobile Rim 2 (NR2)	56.9 km^2	14.0%	9
Peak Near Cabeus B (PCB)	2.1 km^2	0.5%	5
Peak Near Shackleton (PNS)	18.9 km^2	7.4%	6
Slater Plain (SP)	77.0 km^2	19.3%	11
		Total	130

3. Candidate Landing Sites

The sizes of the areas that fulfill the slope and landing accuracy requirements for landing vary significantly among the 16 Artemis CLRs (Table 1). The largest suitable landing area is in the Mons Mouton Plateau (MMP) CLR, which is not surprising as this is by far the largest CLR and, as the name "plateau" implies, a relatively flat surface (Figure 1). Correlating total CLR size with the respective area that meets the HLS requirements, the Slater Plain (SP) region emerges as having the highest fractional area suitable for landing (Figure 2), followed by Nobile Rim 2 (NR2) and Faustini Rim A (Table 1). The smallest suitable landing areas are located in Connecting Ridge Extension (CRE), de Gerlache Rim 1 (dGR1), MMA, and PCB with only a few square kilometers and less than 2.5% of the total CLR area satisfying the constraints for landing (Table 1).

In total, we identified 130 representative CLSs (Table 1) within all 16 Artemis CLRs proposed by NASA (2022a, 2024). All of those CLSs represent the spatial distribution of suitable landing areas (Figure 2) and fulfill the prerequisites defined for the Artemis HLS (NASA, 2019), such as slopes <8° and at least 100 m distance to slopes steeper than 8°. In fact, 125 of the 130 CLSs (~96%) were at slopes shallower than 5°. However, it is important to note that small features, such as young craters or boulders, may have steeper slopes that cannot be resolved at the 20 m/pixel spatial resolution of the DEM. Additional 1- to 2-m scale NAC image analysis should be conducted to evaluate those hazards once the list of potential CLSs has been descoped further.

Because Artemis astronauts will be able to explore up to ~2 km from the selected landing site (NASA, 2019), we prioritized representative CLSs with PSRs within that exploration radius (Figure 2) to support the key science objectives of sampling and understanding volatiles (e.g., National Academies of Sciences, Engineering, and Medicine, 2023; National Research Council, 2007). The CLSs outlined here include potential Artemis landing sites 001, 007, and 011 (associated with Connecting Ridge (CR), Peak Near Shackleton (PNS), and de Gerlache-Kocher Massif (dGKM), respectively; NASA, 2020a), as well as the ones suggested by Kring et al. (2023) labeled dGKM_K2023, HW_K2023, MM_K2023, and NR2_K2023 for the Artemis CLRs dGKM, HW, MMA, and NR2, respectively. The number of CLSs per Artemis CLR ranges from five in MMA and PCB to 11 in HW, MMP, and SP (Table 1). Because the CLSs are manually selected representatives, they can be shifted within the suitable landing area in Figure 2 and still meet the HLS criteria. A list of all 130 landing sites and their coordinates is included in Table A1.

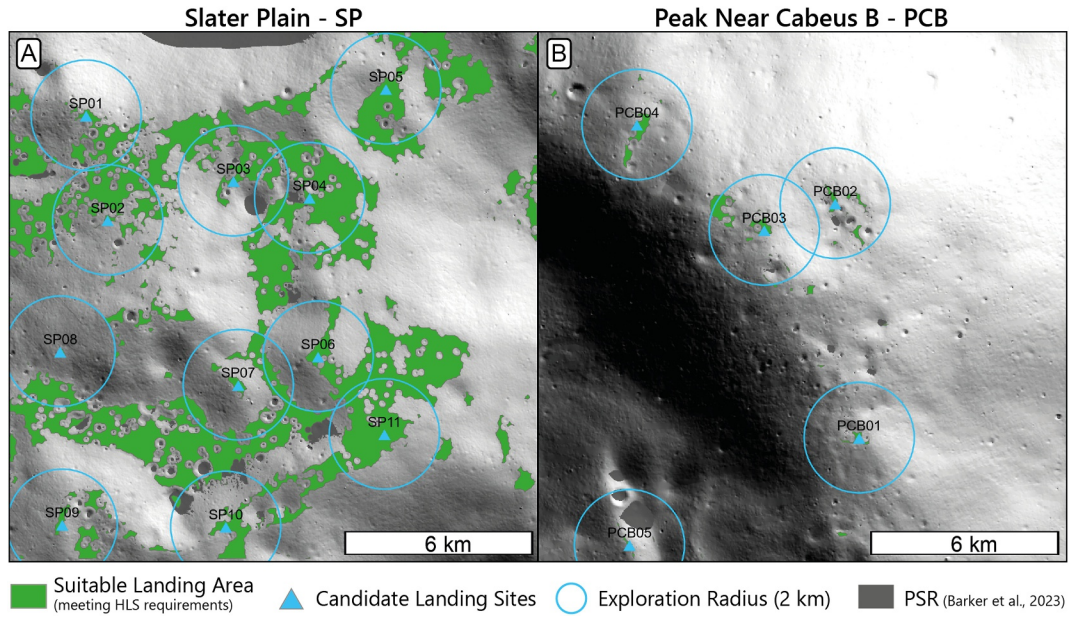


Figure 2. Representative candidate landing sites in the Artemis regions (a) Slater Plain and (b) Peak Near Cabeus B shown as blue triangles superposed on a hillshade map. The green areas represent the suitable landing areas that meet the landing requirements for the Human Landing System (slopes <8° and more than 100 m away from steeper slopes). The blue outline around each landing site represents the maximum radius of 2 km that an astronaut could explore during a walking extravehicular activity (EVA). The coordinates of identified landing sites are included in Table A1 and georeferenced landing sites are available in Wueller et al. (2026).

4. Temporal Variability of Illumination Conditions

Studying the temporal variability of illumination conditions at each CLS is essential for accurate mission planning, as these can affect landing site selection, landing safety, navigation, and the perception of distances and slopes on the lunar surface (e.g., Eppler, 1991; Iqbal et al., 2024; E. M. Jones, 1995; Kring et al., 2025).

4.1. Midsummer Periods From 2025 to 2032

Since the inclination of the Moon's orbital plane with respect to the ecliptic plane changes in a cycle of 18.6 years (Mahanti et al., 2024), lunar seasons do not correspond to consistent calendar dates and change in time and length within the 18.6-year cycle. Table 2 shows the calendar dates for lunar midsummer periods from 2025 until 2032 that we calculated using Planetary and Lunar Ephemeris data (Park et al., 2021; see Section 2.3) and are in agreement with 2025–2029 midsummer dates of Mahanti et al. (2024). The results indicate that peak summer months shift from November in 2025 to July in 2032 (Table 2). After 2032, the calendar dates of the lunar midsummer periods will occur earlier in the year. The longest midsummer of 32 Earth days occurs in 2025 and 2030, whereas the mid-summers in 2028 and 2029 are only 18 Earth days long (Table 2). In certain years (i.e., in 2028), shorter durations occur because many days fall just below the $\leq -1.5°$ subsolar latitude threshold for defining midsummer (Figure 3; e.g., Mahanti et al., 2024). Subsolar latitude versus time highlights the yearly variability and the relationship between subsolar excursions and midsummer duration (Figure 3).

Table 2
Midsummer Periods From 2025 to 2032 (Subsolar Latitude ≤1.5°S)

Year	Start date	End date	Total number of days
2025	November 20	December 21	32
2026	November 06	November 27	22
2027	October 15	November 14	31
2028	September 27	October 14	18
2029	September 15	October 02	18
2030	August 19	September 19	32
2031	August 05	August 26	22
2032	July 13	August 12	31

4.2. Days of Consecutive Illumination

The modeled illumination conditions emphasize that the maximum number of days with consecutive illumination show significant temporal and local variations among the 16 Artemis CLRs (Figure 4; Table A2). The results suggest that the Sun's incidence angle and local topography are the

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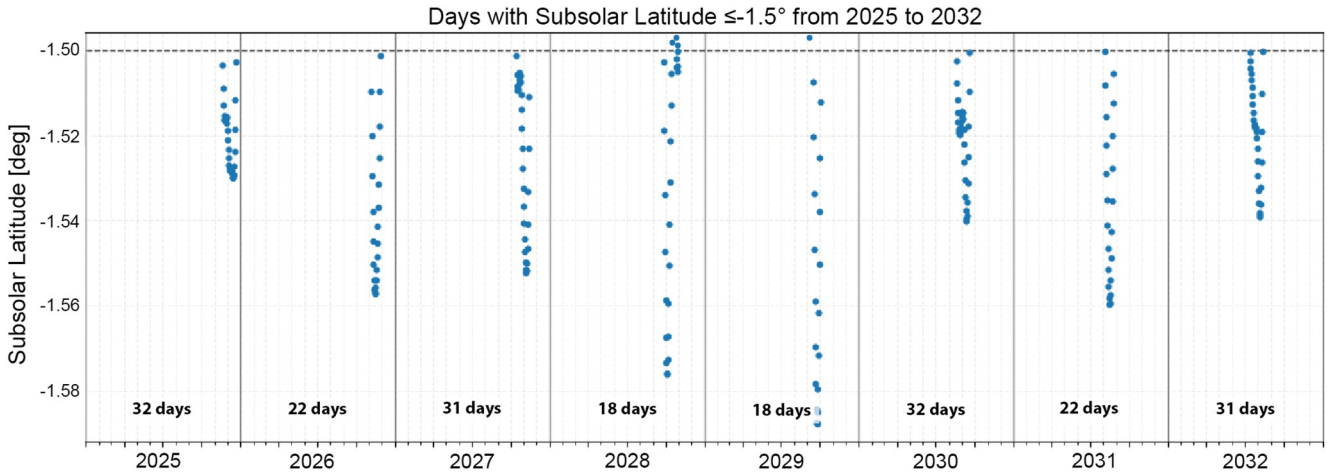


Figure 3. This graph shows subsolar latitude versus time. Each data point represents one day, and the cumulative number of days ≤−1.5° determines the midsummer duration.

strongest controlling factors that determine the illumination at a specific location and time. Because the Artemis III astronauts will stay on the lunar surface for 6.5 days (NASA, 2019), we identify landing sites that are constantly sunlit for the duration of a 6.5-day-long mission to enable solar power generation and to avoid the extremely cold temperatures associated with prolonged periods of darkness. Including a buffer before and after the planned surface mission, we consider illumination for at least 10 consecutive days as favorable for further landing site analysis (Figure 4). Illumination for more than 16 consecutive days would be ideal for longer surface residence and exploration missions. However, the available areas offering these characteristics are extremely limited.

Local and temporal variability in the maximum number of consecutively illuminated days results in some CLSs being available for surface exploration during 1 year but not during other years. For instance, in CR, landing sites with favorable illumination in midsummer 2028 lie in the northern part of the region (Figure 4a), while in midsummer 2030, the southern part has better illumination (Figure 4b). Only one region, located in the center of CR (Figures 4b and 4d; CR06, CR07, and 001), offers consecutive illumination for more than 16 days in all years from 2025 to 2032, making it appealing for missions requiring longer surface resilience time. In contrast, during the midsummer period of 2028, it is predicted that none of the CLSs within Amundsen Rim (AR) shall provide sufficient illumination for a 6.5-day surface stay (Figure 4a). This changes in 2030 when the flat-topped rim segment of Amundsen crater receives sunlight for more consecutive days (Figure 4c), resulting in the availability of most of the AR CLSs for landing and exploration that year.

4.3. Illumination as Constraint for Future Exploration Missions

Long-term illumination modeling demonstrates that landing sites available for landing and surface mobility vary by launch year (Table A2). Nevertheless, many CLSs offer favorable illumination for landing and surface mobility during each midsummer until 2032. Among the 130 CLSs studied here, 97 are available in 2032 and 44 are available in 2031 (Figure 5). Additionally, comparing the change in site availability when considering Sun visibility versus combined Sun and Earth visibility suggests that Earth visibility is not a severely limiting factor in landing site selection. Even if Earth visibility is required for direct communication, 86 CLSs will be available in 2032 and 34 in 2031 (Figure 5). Modeling shows that illumination conditions among the 16 CLRs can differ drastically within a single midsummer. For example, one CLR may have all CLSs available (e.g., AR in midsummer 2026), while others may have no CLSs available (e.g., de Gerlache Rim 2 (dGR2) or PCB in midsummer 2026; Table A2). Similarly, favorable illumination conditions for a CLR can vary from year to year (e.g., MMP is sufficiently sunlit in midsummers 2025, 2029, and 2032, but not in 2026, 2027, and 2031). However, it is important to note that some parts of the CLRs can be available for landing in those years; however, those parts do not have representative CLSs in our study. CLSs that receive sufficient sunlight throughout

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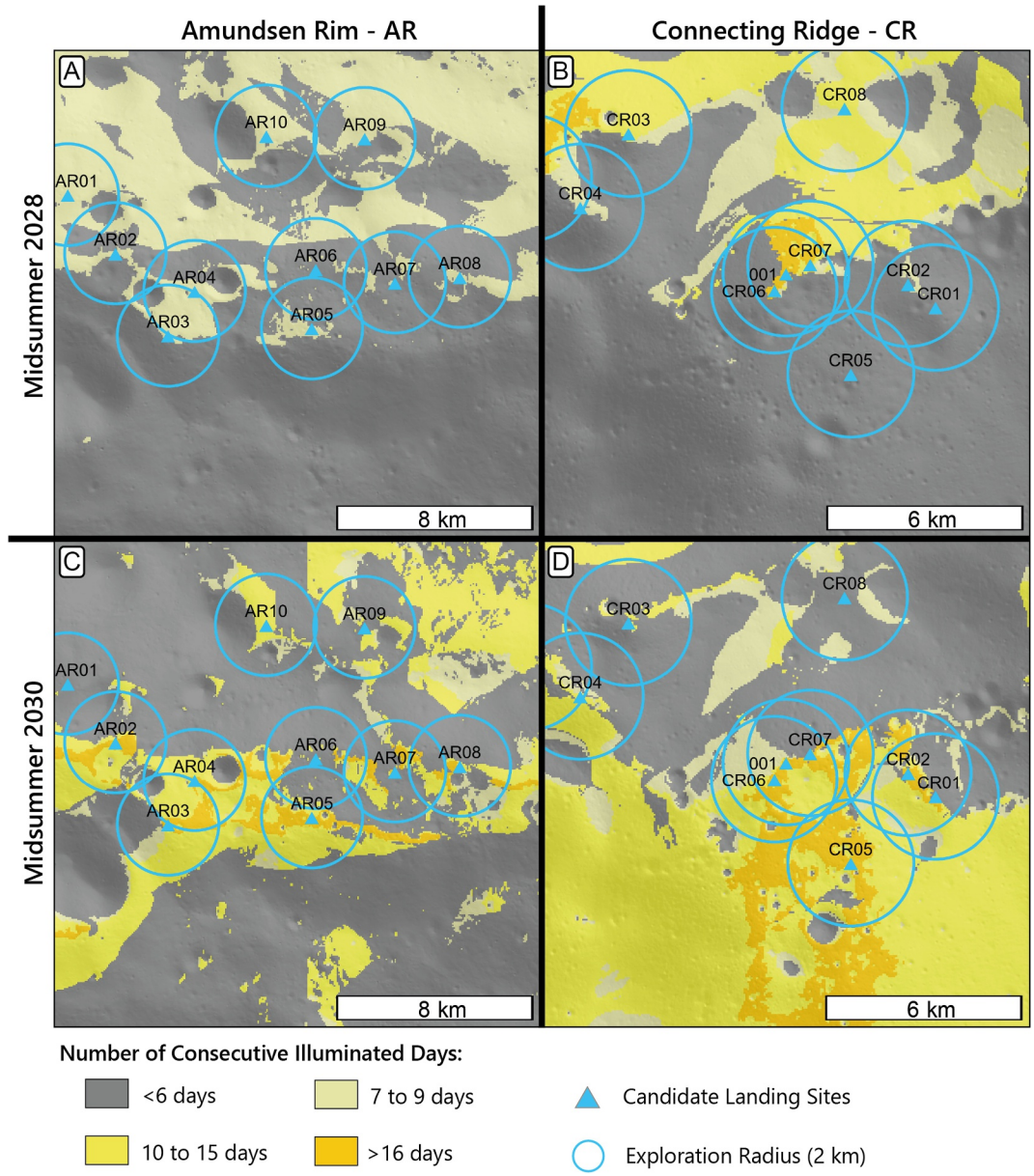


Figure 4. Simultaneous Sun and Earth illumination in the Amundsen Rim region (a) for midsummer 2028 and for (c) midsummer 2030. Illumination conditions in the Connecting Ridge region for (b) midsummer 2028 and for (d) midsummer 2030. The candidate landing sites are shown as blue triangles surrounded by a two-km exploration radius. The exact number of days is included in Table A2 and the georeferenced data are available in Wueller et al. (2026).

midsummer are often located on topographic highs, such as site MMA_K2023 on Mons Malapert or sites MMO01 and MMO02 on Mons Mouton. These CLSs are potential locations for multi-year surface missions, such as a lunar base, as the illumination conditions during midsummer do not vary significantly throughout the years. Testing their potential requires extended illumination modeling for a multi-year period.

In summary, modeling illumination conditions in all 16 CLRs and for all midsummers from 2025 to 2032 shows that landing region and site selection will largely be influenced by the scheduled launch date. Consequently, selecting and ranking multiple landing sites according to their availability for a planned surface mission will provide flexibility in launch times and have backup sites for later mission windows.

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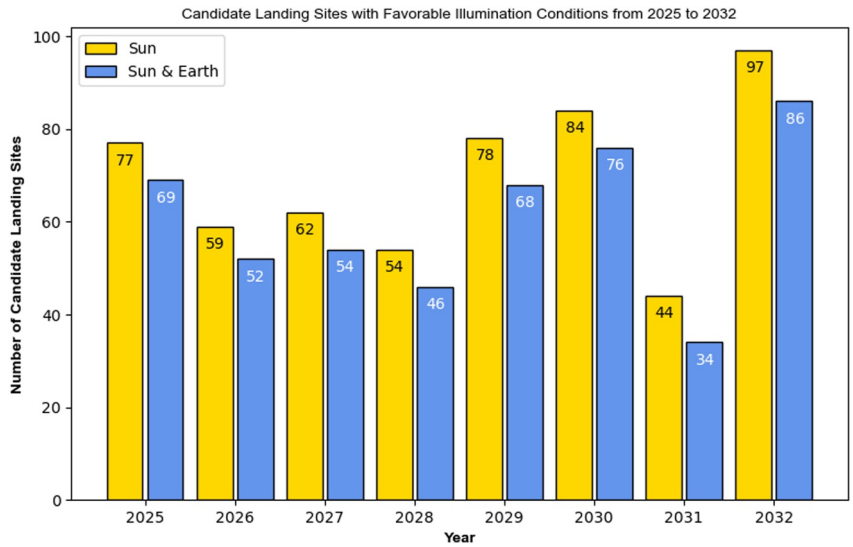


Figure 5. The number of available candidate landing sites (CLSs) out of the 130 CLSs studied that provide at least 10 consecutive days of illumination during the midsummer periods from 2025 to 2032.

5. Identifying the Coldest Accessible Permanently Shadowed Regions

Extreme and temporally variable illumination conditions control surface temperatures in the south polar region. While regions of nearly permanent illumination offer operational advantages, including a more favorable temperature regime for surface exploration, PSRs never receive direct sunlight and can thus maintain cold temperatures. Low temperatures in PSRs may allow for the trapping and accumulation of volatiles, that is, water ice, over geologic timescales (e.g., Vasavada et al., 1999; Watson et al., 1961; Williams et al., 2019), making PSRs

Table 3
List of Candidate Landing Sites That Have the Coldest Accessible PSRs Within a Distance of ~2 km Considering Two Different Threshold Temperatures for Volatile Retention

	Candidate landing sites with accessible PSRs	
Artemis candidate landing region (NASA, 2022a, 2024)	<110 K	110–125 K
Amundsen Rim (AR)	–	–
Connecting Ridge (CR)	01, 03, 04	8
Connecting Ridge Extension (CRE)	–	–
de Gerlache-Kocher Massif (dGKM)	–	–
de Gerlache Rim 1 (dGR1)	103, 104	105
de Gerlache Rim 2 (dGR2)	–	–
Faustini Rim A (FRA)	04, 06, 07	3
Haworth (HW)	10	3
Mons Malapert (MMA)	–	4
Mons Mouton (MMO)	07, 08	–
Mons Mouton Plateau (MMP)	02, 05, 06	03, 08
Nobile Rim 1 (NR1)	–	–
Nobile Rim 2 (NR2)	204	–
Peak Near Cabeus B (PCB)	–	–
Peak Near Shackleton (PNS)	01, 02	–
Slater Plain (SP)	01, 02, 10, 11	04, 08
Total	21	9

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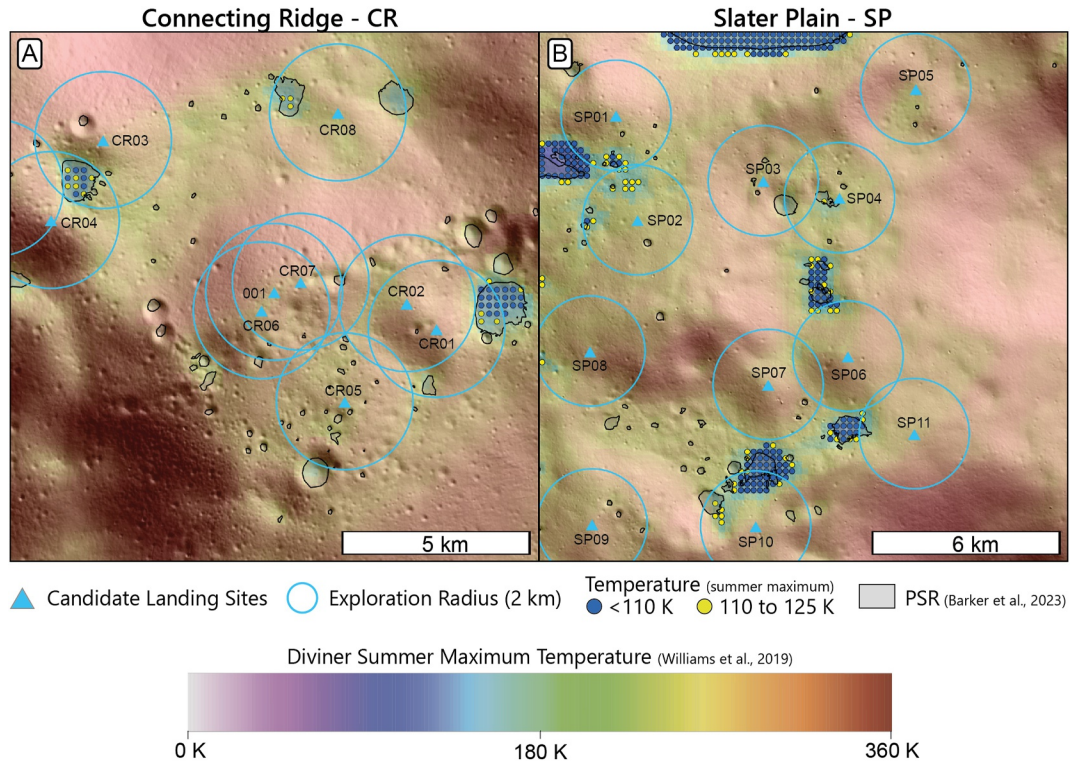


Figure 6. Summer maximum temperature maps (Williams et al., 2019) for Artemis candidate landing regions (a) Connecting Ridge and (b) Slater Plain. Blue and yellow dots indicate areas that are year-round below 110 and 125 K, respectively. These low temperatures usually overlap with Permanently Shadowed Regions and represent potential science targets to study lunar volatile abundance and distribution.

potential resource deposits (Arnold, 1979). If the maximum temperature remains below 110 K throughout the lunar year, volatiles have a high potential to be preserved in PSRs (Vasavada et al., 1999). To identify such areas within our study region, especially near potential landing sites, we used Diviner summer maximum temperature data with a spatial resolution of 240 m/pixel from Williams et al. (2019). Peak summer temperatures are the highest experienced throughout a lunar year; if a PSR sustains temperatures below 110 K during the summer maxima, it likely has favorable conditions to trap and accumulate volatiles during the remainder of the year as well. This will allow us to identify the coldest PSRs that are within the ~2 km exploration radius of our CLSs. These PSRs are considered putative science targets for future surface exploration missions, as they have the highest potential for trapping and preserving volatiles that could be extracted by in situ resource utilization (ISRU) techniques. However, to enable analysis and utilization of lunar volatiles, the PSRs must be accessible, that is, there must be a continuous path from the landing site with slopes never exceeding 15° to reduce metabolic stress for astronauts and to avoid rover slipping (e.g., Iqbal et al., 2024; Scott et al., 1972).

Surface temperatures within the AEZ vary significantly depending on the location and lunar season (Williams et al., 2019). The majority of PSR locations derived from topographic data analyses coincide with surface temperatures less than 110 K. However, not all of the PSRs are cold enough to contain near-surface volatiles, potentially due to secondary illumination, that is, backscattered sunlight from surrounding topography or Earth, affecting local temperatures (e.g., Mahanti et al., 2024). Additionally, PSRs are internally diverse and comprise a spectrum of thermal stability regimes that influence volatile retention, residence times, and chemical composition (Landis et al., 2022).

Twenty-one out of the 130 CLSs outlined in Section 3 have at least one accessible PSR within a 2 km exploration radius that never exceeds 110 K (Table 3). The SP region has four CLSs with accessible and cold PSRs (Figure 6b), while some Artemis CLRs do not offer any landing sites with cold and accessible PSRs, such as AR, CRE, dGKM, dGR2, Nobile Rim 1 (NR1), and PCB (Table 3). If a slightly higher threshold temperature of 125 K for long-term volatile stability is considered, as suggested by Schorghofer and Taylor (2007) and Paige, Siegler,

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et al. (2010), nine more CLSs would provide access to PSRs with a high potential to retain polar volatiles (Table 3), such as landing site CR08 (Figure 6a). All other CLSs that are not included in Table 3 may have PSRs with peak temperatures much warmer than 125 K, suggesting that few to no volatiles were sustained near the surface. However, the burial or superposition of volatiles by an insulating regolith layer may protect volatiles during peak summer temperatures, suggesting that seasonally warmer PSRs can still contain volatiles. Nevertheless, PSRs with lower surface temperature remain more favorable for long-term volatile preservation. Extremely cold PSRs with peak summer temperatures below 90 K are within ~2 km from CLSs FRA04, FRA06, PNS01, PNS02, SP01, and SP10. Consequently, these landing sites and accessible PSRs could be considered prime targets for future surface exploration missions to study volatile abundances and distributions, as per the National Research Council (2007) report's Science Goal 4, and to test ISRU techniques (e.g., Sanders et al., 2008).

6. Identifying the Oldest Accessible Permanently Shadowed Regions

PSR age determination is needed to identify the oldest PSRs that may have accumulated volatiles over extended geologic timescales, potentially preserving a longer and more complete record of volatile delivery and retention processes on the Moon (e.g., Arnold, 1979; M. A. Siegler et al., 2011; M. Siegler et al., 2015; Watson et al., 1961). We inferred PSR-hosting crater ages from the topographic diffusion model (see Section 2.4), which provides the maximum age estimate for the PSR located inside the crater.

6.1. Age Estimates for PSR-Hosting Craters

We focused our diffusion model-based age estimates for PSR-hosting craters on the four Artemis CLRs that are the closest to the south pole: CR, CRE, dGR1, and PNS. In total, we modeled the ages of 115 craters that lie within ~2 km radius of the identified CLSs in these four regions and host accessible PSRs. The craters range in diameters from ~150 m to ~1,400 m, with an average diameter of ~325 m—comparable to terrestrial nuclear test site craters in Nevada (USA), such as Sedan (390 m diameter) and Schooner (295 m diameter) craters, which were used for astronaut training (Kring, 2023). We modeled the erosion of these craters over time using the topographic diffusion model to estimate their ages (see Section 2.4). Of the 115 PSR-hosting craters studied, only two are younger than 100 Ma: a 27 ± 14 Ma old crater with 150 m diameter near CLS CR05 and a 49 ± 28 Ma old crater with 360 m diameter (Figure 7a) on the rim of a larger, degraded impact structure near landing site CR03. The latter is comparable in diameter and age to Cone crater (Apollo 14), which measures 330 m in diameter and is ~39 Ma old (Hiesinger et al., 2015). However, due to steep crater wall slopes (>15°), the ~27,000 m^2 PSR within the 49 ± 28 Ma old crater can only be accessed from the eastern rim, where the slopes are less than 12°. The oldest PSR-hosting crater in the study is located in the PNS region (near CLS PNS03) and has a model age of 3.75 ± 0.04 Ga (Figure 7c). This heavily eroded, ~895 m diameter crater has slopes that never exceed 10°, making its ~96,500 m^2 PSR on the crater floor accessible for future surface missions. Although the diffusion model suggests a wide range of crater ages, the median age is 876 Ma, and 85% of the studied craters are younger than 1,500 Ma (Figure 7d). Only 15% of PSR-hosting craters, typically those with larger diameters, are older than 1.5 Ga.

The CLSs CR05, PNS03, and 007 (both in the PNS region) may provide access to a variety of craters and PSR ages (Figure 8). These sites may have accumulated volatiles for over 2.5 Ga. For instance, landing site CR05 has 15 PSR-hosting craters of different ages, enabling the sampling of several PSRs in different locations to help illuminate our understanding of their thermal stability, and the lateral and vertical distribution of lunar polar volatiles (Science Goal 4a; National Research Council, 2007). The oldest (>2 Ga) PSR-hosting craters—which have the greatest potential to have preserved a complete record of volatile storage, transport, and loss processes (e.g., Arnold, 1979; Siegler et al., 2011, 2015; Watson et al., 1961)—are located in 001, CR02, CR05, CR06, dGR103, 007, PNS02, and PNS03 (Figure 8). To study volatile delivery and retention processes on the Moon (Science Goal 4c; National Research Council, 2007) that were active in the last third of its history, CLSs in the CRE region should be considered, as they host younger PSRs, less than ~1.5 Ga in age (Figure 8). Candidate site dGR2_K2023 has the smallest crater diameter (~200 m) and PSR age range (<500 Ma). The mean age of PSR-hosting craters is statistically highest in PNS and lowest in CRE, with varying numbers of accessible PSR-hosting craters per CLS (Figure 8).

However, it is important to note that the derived crater ages are modeled ages. Small impact craters in the south polar region typically exhibit complex local slopes and lack radial symmetry due to their location on slopes or topographic ridges. Even though the slope can be removed from the present-day crater shape, erosion acts faster on sloped terrain than on flat surfaces (e.g., Aschauer & Kenkmann, 2017; van der Bogert et al., 2018), resulting

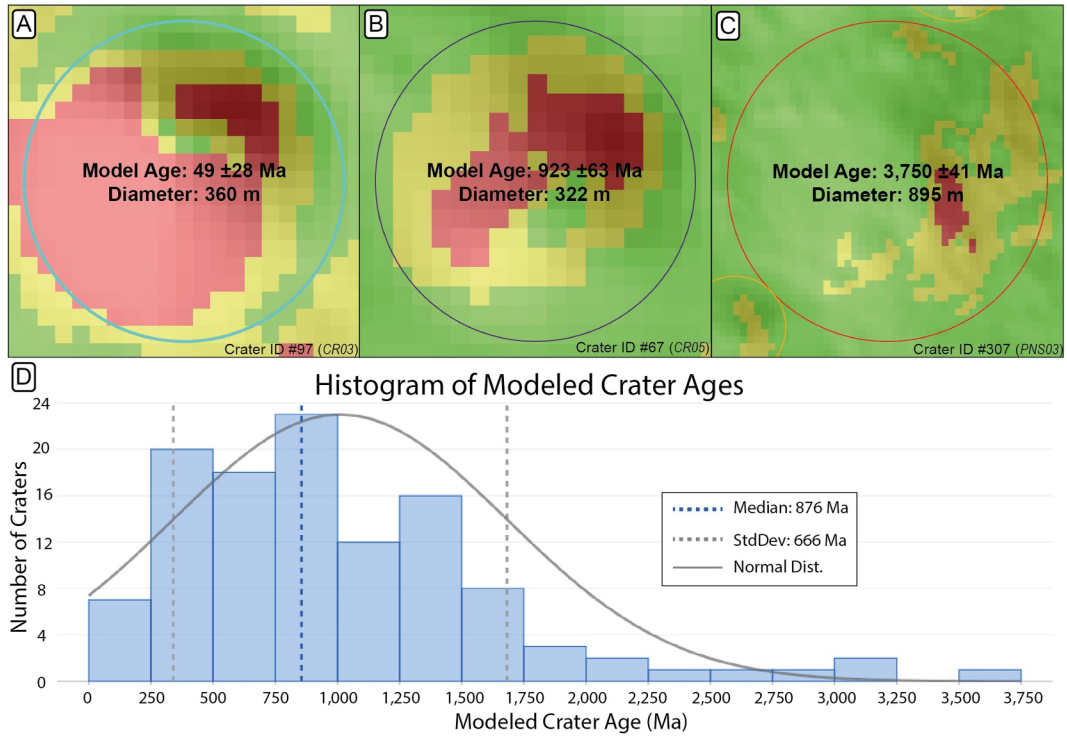


Figure 7. Lunar Orbiter Laser Altimeter slope map (20 m/pixel; Barker et al., 2023) of three craters for which diffusion-based model ages were determined (a–c). Green color indicates shallow surface slopes of less than 10°, yellow shows slopes between 10 and 15°, and red symbolizes steep surface slopes of more than 15°. (d) A histogram of modeled crater ages derived from topographic diffusion modeling.

in ages younger than those modeled by the diffusion model. Furthermore, the model assumes that crater degradation is purely diffusional. Thus, it does not consider impact gardening (e.g., Costello et al., 2021) or non-advective processes, such as landslides and granular flows, which could significantly modify crater topography (e.g., Bickel et al., 2022; Kokelaar et al., 2017; Kumar et al., 2013). Therefore, the model-dependent age estimates should be regarded as the upper bound age of PSR-hosting craters.

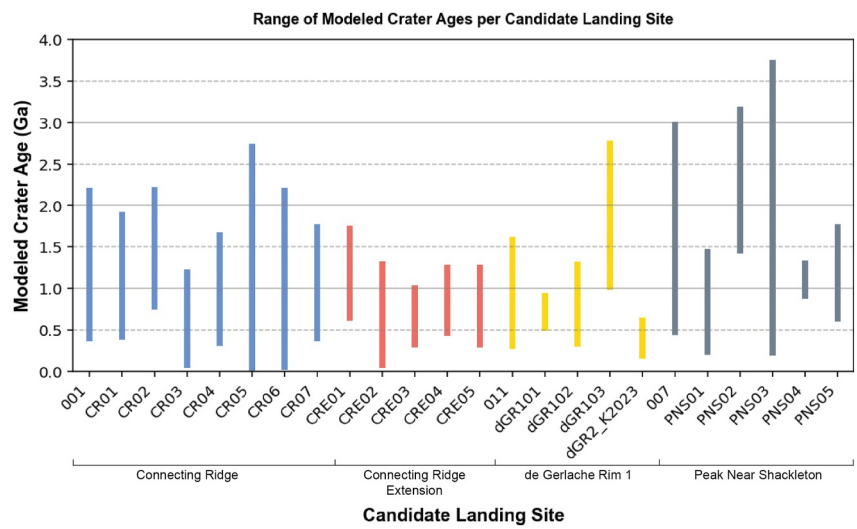


Figure 8. Range of diffusion-based age estimates for PSR-hosting craters within the 2 km exploration radius of the candidate landing sites in Connecting Ridge (CR), Connecting Ridge Extension (CRE), de Gerlache Rim 1 (dGR1), and Peak Near Shackleton (PNS).

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6.2. Processes Delivering and Eroding Lunar Surface Volatiles

Critical information for understanding and quantifying the processes that affect volatiles on the Moon is based on lunar sample analysis, surface temperature measurements, laboratory experiments, or mathematical modeling, and sometimes, some of these data appear contradictory to each other (Hurley et al., 2023; Kleinhenz et al., 2020). Considering that, we present an upper and lower estimate of the volatile content to represent the variations in the literature and the validation that still needs to be done by future missions to the lunar surface.

We identified volatiles (i.e., water) delivered to the lunar surface by solar wind (e.g., Arnold, 1979), micrometeoroids (e.g., Hurley et al., 2017), comets (e.g., Arnold, 1979; Greenwood et al., 2011; Ong et al., 2010; Svetsov & Shuvalov, 2015), and asteroids (e.g., Arnold, 1979; Barnes et al., 2016; Ong et al., 2010; Svetsov & Shuvalov, 2015) as dominant sources of water in the last ∼1.5 Ga. After being trapped in PSRs, volatiles can be released and lost by plasma sputtering induced by solar wind (e.g., Farrell et al., 2013, 2015, 2019; Johnson, 1990), impact vaporization by micrometeoroids (e.g., Cintala, 1992; Farrell et al., 2015, 2019), sublimation (e.g., Andreas, 2007; Arnold, 1979; Schorghofer & Taylor, 2007; Watson et al., 1961), and ballistic sedimentation (e.g., Oberbeck, 1975; Tai Udovicic et al., 2023). We note that endogenic volatiles released by episodic volcanic outgassing events from the Moon's interior were a major volatile source in the past, until approximately 3 billion years ago (e.g., Head et al., 2020; Needham & Kring, 2017; Wilhelms et al., 1987). Younger volcanic activity, though with a much smaller volume, is evident in irregular mare patches (e.g., Braden et al., 2014; Qiao et al., 2020). However, we do not anticipate that the release of volatiles by irregular mare patch activity has significantly contributed to the south polar volatile content because all the PSRs that were studied here are within 2 km of the CLSs in the AEZ, and the irregular mare patches are predominantly located within 30° of latitude of the equator (Braden et al., 2014; Qiao et al., 2020). Consequently, we did not take volcanism-derived inputs into account here.

Solar wind is a high-velocity flow of ions radially emitted by the Sun. As the Moon has neither a global magnetic field nor a thick atmosphere to divert the flow, these ions directly impact the lunar surface (e.g., Hurley et al., 2023). Assuming that the solar wind-implanted hydrogen reduces soil FeO to liberate the oxygen needed to form water, Benna et al. (2019) calculated a solar wind-derived mass flux of 2 g/s H_2O based on the abundance of metallic iron in the lunar soil (Housley et al., 1973). To estimate the amount of water that was successfully transported to and trapped in PSRs, we use a trapping efficiency factor of 0.1 (Schorghofer, 2014) and 0.3 (Ong et al., 2010). Thus, we derive a lower estimate of ∼165 kg_{H2O} km^{-2} Ma^{-1} and an upper estimate of ∼500 kg_{H2O} km^{-2} Ma^{-1} for the contribution of solar wind-derived water to PSRs.

Micrometeoroid impacts are frequent on the Moon (e.g., Gault et al., 1974), and as a few percent of meteorite mass is water (e.g., Hanner & Zolensky, 2010), they are also a polar water contributor. Hurley et al. (2017) estimated a micrometeoroid water flux of 9.46×10^6 kg_{H2O} km^{-2} Ma^{-1}. About ∼30% of the incoming water will get incorporated into impact glass (Daly & Schultz, 2018) and ∼50% of the remaining 70% will be lost to space due to the high vapor temperature (Killen et al., 2018). Therefore, ∼35% of the delivered water would be free to migrate to polar cold traps (Lucey et al., 2022). Of the available water, 10% (Schorghofer, 2014) or 30% (Ong et al., 2010) will be successfully transported and trapped in PSRs, suggesting a micrometeoroid-delivered water influx of ∼175 kg_{H2O} km^{-2} Ma^{-1} to ∼524 kg_{H2O} km^{-2} Ma^{-1}, respectively.

Asteroid and comet impacts are not constant over time, leading to a declined impact mass rate in the last billion years compared to the early bombardment history of the Moon (e.g., Stoffler et al., 2006). Mathematical modeling by Ong et al. (2010) found that ∼6.5% of the water mass delivered to the Moon by comets is retained, which is equivalent to 1.2×10^{10} to 3.9×10^{11} metric tons of water delivered over the last billion years. Applying the effects of migration and survival for water molecules (Butler, 1997; Crider & Vondrak, 2003), ∼1.1% of the total water mass delivered to the Moon can get trapped in PSRs (Ong et al., 2010). Based on these calculations, we estimate a water influx of 3.4–113 kg_{H2O} km^{-2} Ma^{-1} into south polar PSRs from cometary impacts. Furthermore, Ong et al. (2010) estimated that ∼16.5% of the asteroidal mass impacting the Moon is retained. Estimates of asteroid water content vary by types: carbonaceous chondrites can have up to 20% water (Baker, 1971), whereas ordinary chondrites rarely contain water (Mason, 1972). Water accounts for ∼2.2% of the mass of impacting asteroids, assuming that one-third of them are C-type asteroids (Shoemaker et al., 1990) and two-thirds of C-type asteroids contain water (T. D. Jones et al., 1990), and they have an average water content of 10% by weight (typical for CM meteorites; Rivkin et al., 2003). Using impactor fluxes from Shoemaker et al. (1990) and Werner et al. (2002) and the average water content of 2.2 wt.%, Ong et al. (2010) calculated that 2.5×10^{15} kg of water

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was delivered to the Moon in the last 1 Ga. Applying migration and survival rates (Butler, 1997; Crider & Vondrak, 2003), we estimated a water influx into south polar PSRs due to asteroidal impacts of ~712 kg_{H2O} km^{-2} Ma^{-1}.

Plasma sputtering by solar wind is primarily independent of the surface temperature (Farrell et al., 2019) and can effectively erode surface volatiles in the uppermost few nanometers (Johnson, 1990). The ice composition ratio of the ice-regolith mixture controls the sputtering rate of water, yielding a water release rate of 8×10^7 H_2O m^{-2} s^{-1} for a 1 wt.% ice-regolith mixture (Farrell et al., 2019), equivalent to ~75.6 kg_{H2O} km^{-2} Ma^{-1}. Most of the released water molecules do not have the velocity (>2.4 km/s) necessary to escape lunar gravity and thus will land elsewhere on the Moon (Farrell et al., 2013, 2015). Consequently, the estimate used here is likely a higher estimate of released water as local topography can capture liberated water molecules excavated in highly oblique emission angles (Farrell et al., 2019).

Impact Vaporization due to micrometeoroids can continuously release surface volatiles trapped in PSRs (e.g., Farrell et al., 2019). Impact vaporization only erodes surficial volatiles, as micrometeoroids only penetrate <5 mm into the surface (Farrell et al., 2013) and have impact energies too low to emit large vapor volumes. Using the vaporization mass flux of Cintala (1992) and a 1 wt.% icy regolith, Farrell et al. (2019) suggested a volatile release rate of 2×10^8 H_2O m^{-2} s^{-1}. Therefore, we calculated that the water release rate from south polar PSRs is ~189 kg_{H2O} km^{-2} Ma^{-1} assuming that the icy regolith contains 1 wt.% of ice. However, impact vaporization is a lower energy process compared to sputtering and thus, even less molecules reach escape velocities (Farrell et al., 2015). Consequently, our estimate represents a higher limit for the lost volatiles, as many of the emitted water molecules are repeatedly available for migration and trapping in PSRs.

Sublimation driven by temperature can substantially release volatiles from the lunar surface (e.g., Andreas, 2007; Arnold, 1979; Schorghofer & Taylor, 2007; Watson et al., 1961). If a PSR's temperature increases by 10 K from 110 to 120 K, the sublimation rate increases by a factor of 100 (Andreas, 2007), highlighting an extreme temperature sensitivity (Figure S2.1 in Supporting Information S1). To infer the thermal environment of each PSR within the ~2 km exploration zones of the CLSs, we calculated the average temperature for a lunar year using Diviner temperature maps from Williams et al. (2019). The sublimation rate depends on temperature, ice-regolith ratio, regolith grain size distribution, and the physical form of water ice such as pure water ice deposits on the surface, ice coating regolith grains with convex ice surfaces, or ice connecting two grains with concave ice surfaces (Figure S2.2 in Supporting Information S1). At 40 K, the sublimation rate of a convex or concave surface (with a radius of 5 μm) versus a planar surface is ~2.3% higher and ~2.3% lower, respectively. At 100 K, the difference in sublimation is less than 0.8% between a convex and a concave surface with respect to a planar surface (Figure S2.3 in Supporting Information S1). At higher temperatures and larger radii of curvature (larger grains) the difference between sublimation from a curved and planar surface decreases rapidly. For an average regolith grain size of 70 μm, as sampled by Apollo 16 (Carrier et al., 1991), and a temperature of 110 K, the differences in sublimation are of the order of 1.35×10^{-5} and thus below the uncertainty introduced by the Diviner temperature data. Therefore, we used the sublimation rate for planar ice surfaces to estimate the total volatile content of PSRs with 1 wt.% icy regolith. See Text S2 in Supporting Information S1 for the equations and a detailed description of how we calculated the sublimation rate for volatiles within PSRs.

Ballistic sedimentation describes the process of ejecta following ballistic trajectories at high velocities and mixing ejecta material with local materials by impacting the surface, causing heating of the mixed ejecta (Oberbeck, 1975). Disruption of south polar volatiles in PSRs by ballistic sedimentation was modeled by Tai Udovicic et al. (2023) for craters larger than 20 km in diameter. They inferred the threshold kinetic energy for the onset of ballistic sedimentation on Earth to be between 10 and 1,500 MJ/m^2 (Tai Udovicic et al., 2023). Additionally, the deposition of continuous ejecta is restricted to within 4 crater radii from the crater center (Melosh, 1989). However, since the PSR-hosting craters in this study are relatively young (75% of the studied craters are younger than 1,332 Ma) and small (<1 km in diameter), none of them reached the kinetic threshold energy required to initiate ballistic sedimentation. Larger craters in the vicinity (e.g., Spudis or Marvin craters near CLRs CR and CRE) are more degraded and thus likely older than the small craters hosting PSRs close to the CLSs. Consequently, those larger craters formed prior to the PSRs and could not have disrupted the volatile deposits.

6.3. Estimated Volatile Volumes

To estimate potential volatile abundances in south polar PSRs in kilograms (kg), we calculated the volatile influx and loss rate for a given area and time period (in kg km^{-2} Ma^{-1}) and multiplied the difference of both with the modeled age of the PSR-hosting crater (in Ma; Thaker et al., 2025). This provides information on whether a PSR has the potential to trap and accumulate volatiles over time. For each CLS in the four Artemis CLRs (CR, CRE, dGR1, and PNS), we estimated the cumulative volatile abundance of all PSRs within craters 2 km from the landing site for which we have model ages (Figure 9). These estimates take the variation in the literature on the volatile in- and outflux rate and temperature-dependent sublimation rate into account (see Section 6.2), providing upper and lower estimates of volatile abundance. These calculations reflect volatile delivery and loss rates in the literature (Andreas, 2007; Arnold, 1979; Barnes et al., 2016; Cintala, 1992; Farrell et al., 2013, 2015, 2019; Greenwood et al., 2011; Hurley et al., 2017; Johnson, 1990; Oberbeck, 1975; Ong et al., 2010; Schorghofer & Taylor, 2007; Svetsov & Shuvalov, 2015; Tai Udovicic et al., 2023; Watson et al., 1961), producing modeled potential volatile abundances within CLS PSRs. While the calculations are not prima facie evidence for the presence of volatiles, they can be used to guide the selection of landing sites that can be used to test delivery and loss rates and thus improve such calculations for future mission activities.

Due to the limited data on the storage depth and subsurface stability of lunar volatiles, we simplified our calculation by assuming that volatile reservoirs are at the PSR surface. However, if the ice is buried beneath a thin regolith layer, it would be protected from plasma sputtering and impact vaporization. In addition, if the ice is not exposed at the surface, an insulating layer of regolith would protect it from extreme surface temperatures, significantly reducing volatile loss by sublimation. Consequently, the quantification of ice loss processes may be an overestimation of volatiles released from PSRs in the study area. Therefore, all volatile estimates presented here should be considered approximations showing whether PSRs have the potential to preserve trapped volatiles rather than absolute values. In addition, landing and surface operations within a few kilometers of PSRs may locally perturb surface thermal conditions and volatile stability, requiring consideration of environmental sensitivity for future missions (Ahrens et al., 2025). To gain a more complete understanding of the processes affecting lunar volatiles, ground truth verification by future exploration missions is essential. Such missions would provide direct observations and measurements to validate existing models and refine our knowledge of volatile behavior on the Moon.

Of the 27 CLSs identified in the four Artemis CLRs for which we have volatile estimates, only 14 have PSRs that are cold and old enough to trap and accumulate volatiles over time. The cumulative volatile abundance of these landing sites can differ significantly, ranging from 409 kg in CRE02 to nearly 644,952 kg in PNS03 (Figure 9). Generally, older and/or colder PSRs have the most volatiles because they have had the most time to accumulate and/or preserve volatiles, as sublimation is less effective at low temperatures (e.g., Andreas, 2007; Arnold, 1979; Schorghofer & Taylor, 2007; Watson et al., 1961; see Section 6.2). The distinct correlation between average PSR temperature and estimated volatile abundance suggests that temperature is the driving factor for volatile accumulation. ISRU of polar volatiles is part of the Artemis program's "sustainable exploration" goal (NASA, 2020a). As reflected in NASA's Moon to Mars strategy (NASA, 2022b), the water can be used as is or to produce oxygen and hydrogen to fuel vehicles, power systems, and life support consumables. If the objective is the production of 10 tons of oxygen, approximately 15 tons of water per year are required (Sanders et al., 2008; Sanders & Kleinhenz, 2022). However, when considering an icy regolith containing ∼1 wt.% of water, it should be noted that about 1,500 truckloads of lunar regolith (each truckload equaling one metric ton) will be needed to extract 15 tons of water, requiring a large, well-organized infrastructure. Although the Artemis III mission will not rely on ISRU, and will lack the necessary infrastructure (Kleinhenz & Sanders, 2023), ISRU will become increasingly important for subsequent Artemis missions and other future surface exploration activities (NASA, 2022b). Most CLSs could sustain the 15-ton objective for at least one year (Figure 9). Landing sites in the CRE region could support ISRU for lunar infrastructure for a maximum of 2 years (assuming the upper volatile estimate). Landing site CR05 could potentially fulfill the ISRU objective for approximately 13 years under a conservative scenario based on the lower volatile estimate, or ∼28 years under a more optimistic scenario based on the upper estimate. Only landing site PNS03, if the presence of water is confirmed there, has the potential to meet the goal of 15 tons per year for 20 to 43 consecutive years.

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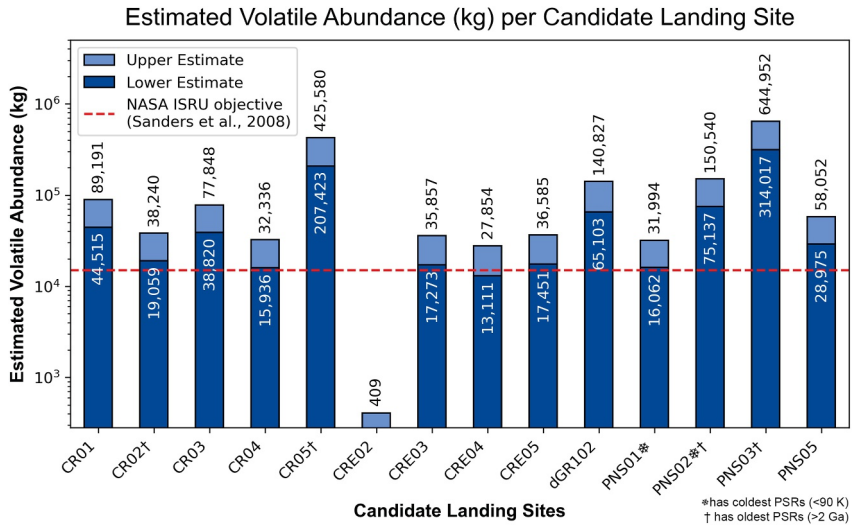
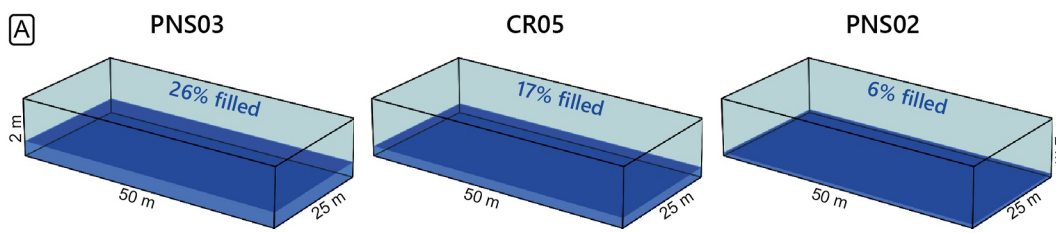


Figure 9. Estimated volatile abundance for permanently shadowed regions within 2 km of candidate landing sites in Connecting Ridge (CR), Connecting Ridge Extension (CRE), de Gerlache Rim 1, and Peak Near Shackleton. The upper and lower volatile estimates reflect the uncertainty in the literature regarding quantifying the flux of volatiles being delivered or eroded. The dashed red line indicates the tonnage required per year to fulfill NASA's in situ resource utilization objective of 15 metric tons (Sanders et al., 2008).

To provide a different perspective, we compared the estimated volatile volumes to the volume of an Olympic swimming pool (50 × 25 × 2 m). This illustrates that even the highest cumulative volatile estimates in PNS03, CR05, or PNS02 could only fill an Olympic pool by ∼26%, ∼17%, or ∼6%, respectively (Figure 10a). However, the water ice on the Moon is incorporated into the regolith and extends over a much larger area. Therefore, the estimated volatile volumes are distributed among several PSRs no more than ∼2 km from the landing sites and are not concentrated in one location. For long-term ISRU, this would require relocating technical equipment to several PSRs, which could be kilometers apart. For a longer surface mission that depends on ISRU, larger PSRs more than ∼2 km from the landing site zones can be considered as they are tangentially related to older craters and have lower, more stable temperatures. Consequently, they likely have a higher potential to harbor more volatiles, as modeled in Tai Udovicic et al. (2023) and Lucey et al. (2022), which suggests volatile retention under pro-

A



B

	2025		2026		2027		2028		2029		2030		2031		2032	
	Sun	Sun & Earth	Sun	Sun & Earth	Sun	Sun & Earth	Sun	Sun & Earth	Sun	Sun & Earth	Sun	Sun & Earth	Sun	Sun & Earth	Sun	Sun & Earth
CR05	12	11	15	15	18	18	3	3	6	4	18	18	0	0	6	6
PNS02	5	4	8	8	4	3	11	9	11	10	14	11	2	1	14	12

Figure 10. (a) Comparison of the cumulative volatile estimates for landing sites PNS03, CR05, and PNS02 is shown in relation to the volume of an Olympic swimming pool (50 × 25 × 2 m), with a vertical exaggeration of five. Note that volatiles in permanently shadowed regions are spread over extensive areas and not exist as ice blocks. (b) The maximum number of consecutive days with favorable illumination condition (>10 days) for landing sites CR05 and PNS02, highlighted in green.

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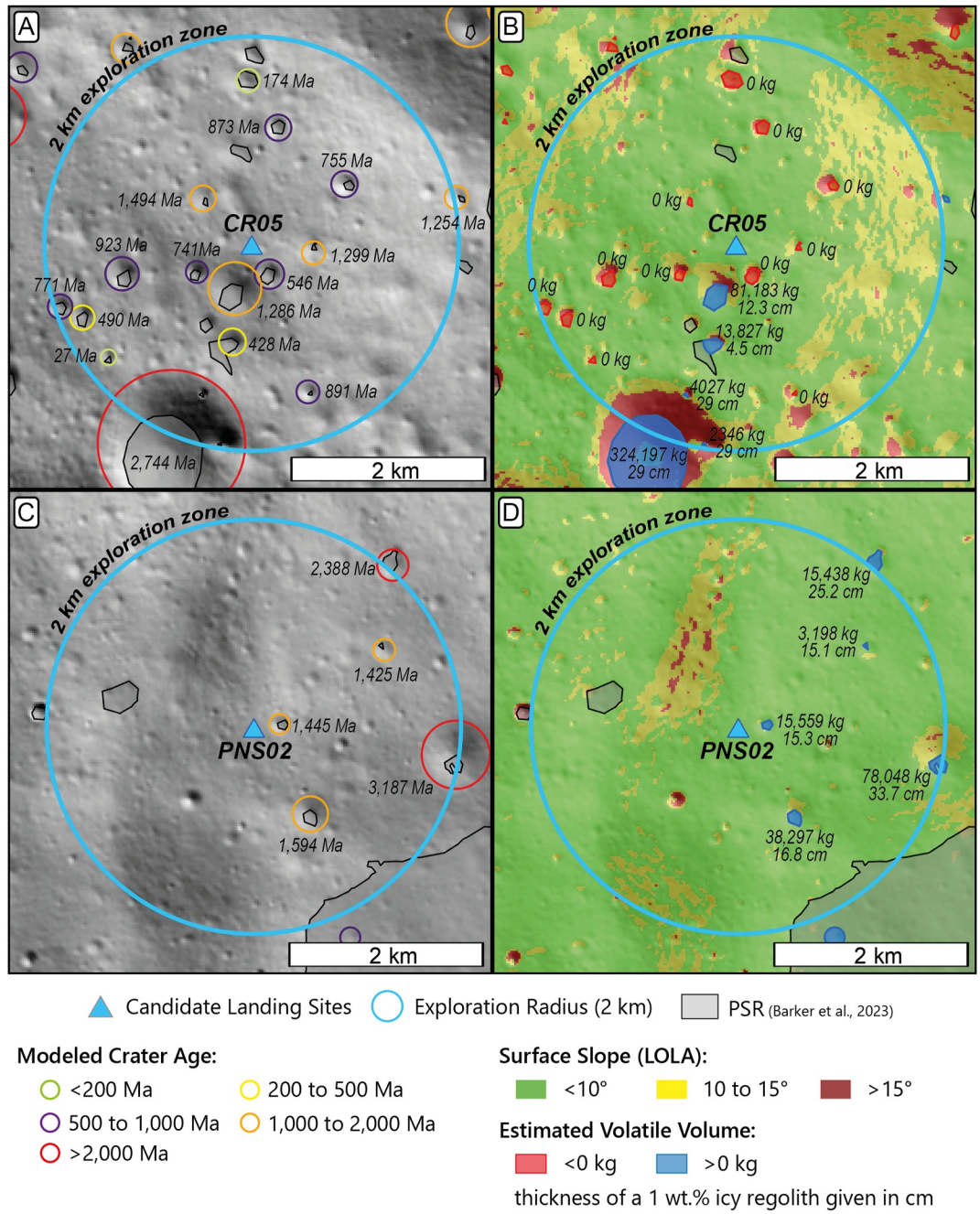


Figure 11. The distribution and modeled formation age of PSR-hosting craters within ~2 km of the candidate landing sites (a) CR05 and (c) PNS02 are displayed. Panels (b and d) show the estimated abundances of volatiles in permanently shadowed regions and the thickness of a putative icy regolith layer containing 1 wt.% water ice superposed on a 20 m/pixel Lunar Orbiter Laser Altimeter slope map.

tective ejecta deposits from younger impacts. However, these younger ejecta deposits can reach substantial thicknesses, necessitating drilling to reach the buried volatiles.

Next, we focus on two CLSs–CR05 and PNS02–that potentially have the highest volatile abundances within a ~2 km radius (Figure 9). Candidate site CR05 (89.421°S, 149.472°W) is ~8 km away from the rim of Shackleton crater in the CR region. The landing site is sufficiently illuminated for exploration in the midsummers of 2025, 2026, 2027, and 2030 (Figure 10b). The area surrounding the landing site contains 15 PSR-hosting craters (Figure 11a) ranging in diameter from ~150 m to ~1.4 km. The smallest crater is the

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youngest, 27 ± 14 Ma, while the largest crater is the oldest at ∼2.74 ± 0.02 Ga (Figure 11a). This variety in crater diameter and model age enables studies of temporal variability in crater degradation processes and processes affecting the lunar volatile distribution, addressing key scientific objectives (e.g., National Research Council, 2007) and providing vital ground truth to test diffusion and thermal models. However, our volatile estimates suggest that most of the PSRs in CR05 for which we have ages were unable to maintain surface volatiles (Figure 11b). Nevertheless, two PSRs close to the landing site (within 1 km) and one farther away potentially have trapped and preserved ∼425,000 kg of volatiles over time (Figure 9), making them prime targets to study the lunar water cycle, the characteristics of the icy regolith, and the processes involved in trapping and preserving volatiles in PSRs. The two volatile-bearing PSRs closest to CR05's landing site can be accessed via slopes shallower than 10° to reach the edges and <15° to go deeper into the PSRs (Figure 11b). The largest PSR, located in a ∼1.4 km diameter crater, contains the most potential volatile mass but can only be accessed from the northwestern rim, requiring longer traverses or additional roving equipment. Assuming an ice-regolith ratio of 1 wt.% and even distribution of volatiles across the PSR area, the icy regolith in the three volatile-rich PSRs would be between 4.5 and 29 cm thick (Figure 11b).

Candidate landing site PNS02 (89.061°S, 128.442°E) is located between Shackleton and Slater craters, near the southern border of the PNS region, and is available in the midsummers of 2028, 2029, 2030, and 2032 (Figure 10b). PNS02's exploration zone contains five PSR-hosting craters that vary in diameter from 190 to 660 m and in model age from ∼1.45 to 3.2 Ga (Figure 11c). These five craters potentially preserve volatiles (Figure 11d), with a cumulative abundance of ∼150,000 kg (Figure 9). The five PSRs are accessible with slopes never exceeding 10° (Figure 11d), have generally low yearly average temperatures of 62–91 K, and have a high potential to have preserved significant amounts of volatiles in a 1 wt.% icy regolith layer varying in thickness from 15.1 to 33.7 cm (Figure 11d). Assuming a 5.6 wt.% abundance of water ice in the regolith, as suggested by the LCROSS Centaur impact experiment in Cabeus crater (Colaprete et al., 2010), the thickness of the putative icy regolith would range from 0.8 to 5.2 cm. Only 250 m from the landing site is a ∼1.45 Ga old, 6,800 m^2-large, and 91 K cold PSR with accessible slopes, which is estimated to sustain ∼7,750 to 15,550 kg of water ice (lower and upper estimates, Figure 11d). In addition to the PSRs for which we have model ages, there is a 5.9 km^2-sized PSR located within a ∼5 km diameter degraded impact crater north of the PNS02 landing site (Figure 11d). The margin of this extensive PSR could be accessed within a distance of ∼2 km from the landing site by traversing slopes shallower than 10°. This PSR is likely older than those studied, and Diviner temperature data (Williams et al., 2019) suggest that peak summer temperatures remain below 110 K (Figure 11d). The low temperatures suggest that chemically diverse volatile species (Landis et al., 2022) may be preserved and available for sampling in future exploration endeavors.

7. Considerations for Future Exploration

We identified CLSs that meet the HLS requirements for all 16 Artemis CLRs. Additionally, we modeled the illumination for the midsummer periods from 2025 to 2032 to determine the maximum number of consecutive illuminated days that limits site availability depending on the launch dates. Furthermore, we investigated the highest temperatures experienced by the PSRs close to the landing sites to evaluate their ability to retain volatiles. For four CLRs (CR, CRE, dGR1, and PNS), we derived diffusion model ages for PSR-hosting craters accessible from the landing sites to estimate volatile abundance in these PSRs, considering processes that deliver and erode surface volatiles on the Moon. We summarize the results and data presented in this work, as well as their implications for the future exploration of each of the 16 Artemis CLRs in this section. However, it should be noted that suggestions regarding individual sites or regions are based on the distribution of potentially interesting PSRs within 2 km of HLS-compatible landing sites and with nominal <20° slopes suitable for walking EVA; they do not yet include an assessment of other risks and hazards (e.g., Carton et al., 2025; Jo et al., 2024; Lee et al., 2025).

Amundsen Rim (AR) is located on the western rim of the ∼100 km diameter Amundsen crater (Figure 1; NASA, 2022a), which was created in the Nectarian period ∼4.04 Ga ago (Wueller et al., 2024). The landing region exhibits a high relief terrain, including parts of the inner crater wall, a flat-topped rim segment, and parts of the Amundsen ejecta blanket. Ten representative landing sites were identified that fulfil the HLS

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requirements of less than 8° surface slope and a distance of more than 100 m from steeper slopes (NASA, 2019). Illumination conditions are favorable during the midsummers of 2026 and 2029, when all sites are accessible for at least 10 consecutive days. In contrast, no sites are suitable during the midsummers of 2027, 2028, and 2032. CLSs AR02, AR03, AR04, and AR08 offer recurring availability through 2032. Peak PSR temperatures can exceed 125 K in the summer, yet volatiles may accumulate seasonally or have been buried beneath insulating regolith.

Connecting Ridge (CR) was included in NASA's (2022a) list of CLRs but was excluded from the NASA (2024) update. It is situated near the boundary between the lunar nearside and farside and includes the southern part of the Shackleton-Henson-de Gerlache Ridge (Figure 1). The terrain was likely buried by ejecta from the Shackleton impact ∼3.3 Ga ago (Kring et al., 2021). We identified nine representative landing sites that meet the HLS requirements, including the previously suggested site 001 (NASA, 2020a). While illumination varies from year to year, sites CR05, CR06, CR07, and 001 remain sunlit for over 10 days during midsummer until 2032. This makes them promising locations for extended surface operations. Most landing sites show a minor difference between Sun-only and combined Sun-Earth visibility, though some exceptions occur (Table A2). Several sites, notably CR01, CR03, CR04, and CR08, provide access to PSRs with extremely low temperatures, enabling volatile accumulation and potentially substantial volatile deposits in these regions. Landing site CR05 has both the oldest and youngest PSR-bearing craters in the area with a cumulative potential of ∼425,000 kg of volatiles (see Section 6.3). This amount is sufficient to support NASA's ISRU target of 15 tons per year for multiple years (Sanders et al., 2008; Sanders & Kleinhenz, 2022).

Connecting Ridge Extension (CRE) lies north of the CR region, along the northern part of the Shackleton-Henson-de Gerlache Ridge, near the ∼20 km diameter de Gerlache crater (Figure 1; NASA, 2022a). Its steep terrain limits accessible landing areas, with only 1.9% meeting the slope criteria. Nevertheless, six representative landing sites have been outlined and exhibit varying illumination conditions until 2032. For instance, during the midsummers of 2027 and from 2030 to 3032, most of the landing sites offer suitable illumination for a surface mission. However, in other years, only a few or none of the sites are available for a 6.5-day surface mission. At site CRE01, the illumination conditions are excellent throughout all years until 2032. Some locations, such as CRE06, are only suitable for short missions due to limited sunlight. All PSR-hosting craters within ∼2 km of the landing sites have model ages of less than 2 Ga, with most being younger than 1 Ga. PSRs in CRE02 to CRE06 were able to maintain volatiles, although the cumulative volatile estimate for CRE02 is 409 kg, while the other sites have an estimated cumulative volatile abundance ranging from 28,000 to 36,500 kg.

De Gerlache-Kocher Massif (dGKM) is located on the lunar far side atop Mons Kocher (Figure 1; NASA, 2022a). Seven CLSs were identified, including the previously suggested site, dGKM_K2023 (Kring et al., 2023). In the midsummers of 2026, 2027, and 2030 to 2032, most sites receive sunlight for more than 10 consecutive days. However, some locations never exceed this threshold but can still support shorter missions, such as dGKM04 and dGKM06. Summer temperatures in nearby PSRs exceed 125 K; however, volatiles may remain protected in the subsurface.

De Gerlache Rim 1 (dGR1) lies along the eastern rim of de Gerlache crater and partially overlaps the dGR2 candidate region (Figure 1; NASA, 2022a, 2024; Kring et al., 2020a, 2020b). Six representative landing sites that meet the HLS requirements have been identified, including the previously suggested (NASA, 2020a) and characterized (Kumari et al., 2022) site 011, which has favorable illumination throughout all midsummers until 2032. Although illumination varies significantly, at least one candidate site remains accessible for landing and surface exploration every year. Several sites, such as dGR103 and dGR104, provide access to PSRs with temperatures below 110 K and should be considered targets for analyzing processes affecting polar volatile distribution. However, only dGR102 shows significant volatile retention, with an estimated ∼140,000 kg near the surface.

De Gerlache Rim 2 (dGR2) is located just north of dGR1 and overlaps with the same region (NASA, 2022a). We selected seven landing sites representative of the spatial variability of suitable landing areas, including the previously proposed site dGR2_K2023 (Kring et al., 2023). This site offers excellent illumination and will maintain more than 14 consecutive days of Sun and Earth visibility from 2027 onward. During the midsummers of 2027 and from 2030 to 2032, most of the seven CLSs will support extended surface missions. Although peak PSR temperatures exceed 125 K during southern summers, buried volatiles may be preserved under protective layers

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that minimize volatile loss due to plasma sputtering caused by the solar wind, vaporization from micrometeoroid impacts, and sublimation.

Faustini Rim A (FRA) is positioned near the boundary between the lunar nearside and far side, near the southern rim of the Faustini crater (Figure 1; NASA, 2022a). We identified eight CLSs that represent the size and distribution of area that meets the HLS requirements for landing, with at least a few sites consistently offering suitable illumination for surface stays of at least 6.5 days. Candidate sites FRA04 and FRA06 provide access to extremely cold PSRs, with temperatures below 90 K in midsummer, making them prime targets for studying volatiles and testing ISRU techniques.

Haworth (HW) is located near the rims of the pre-Nectarian Haworth and Cabeus craters (Figure 1; NASA, 2022a, 2024). We identified 11 potential landing sites, including site HW_K2023 (Kring et al., 2023). While the illumination conditions vary, a few landing sites are available each year for a 6.5-day surface mission, as planned for Artemis III (NASA, 2019). Candidate sites HW04 and HW10 offer extended Sun and Earth visibility exceeding 25 days, making them attractive for longer surface missions. Additionally, from landing site HW10 PSRs with temperatures below 110 K are accessible, presenting opportunities to study the thermal environment and potentially sample volatiles.

Mons Malapert (MMA), formerly known as Malapert Massif, is situated on a ∼6 km high massif that is interpreted to be the remnant rim of the SPA basin (e.g., Bernhardt et al., 2025). Five CLSs are distributed along the massif's ridge and southern flank. Sites on the crest receive favorable illumination during the midsummers of 2025,2028, 2029, and 2032. Meanwhile, site MMA04, located on the southern base, provides limited but adequate sunlight for 6.5-day surface missions in 2027 and from 2030 to 2032. Furthermore, within ∼2 km of landing site MMA04, PSRs with temperatures below 125 K are accessible, making it the most promising site in the region for volatile exploration. PSRs on the ridge of Mons Malapert receive more sunlight and are expected to experience warmer conditions, reducing their potential for preserving surface volatiles.

Mons Mouton (MMO), originally outlined as Leibnitz Beta Plateau in NASA (2022a) and renamed in NASA's (2024) update, is located near the southern margin of the ∼8 km high Mons Mouton. This region overlaps with the other Artemis CLRs, MMP and NR1 (Figure 1). We identified 10 CLSs in the region, most of which will have suitable illumination conditions during the midsummers of 2025, 2028, 2029, and 2032. However, during the remaining midsummers until 2032, significantly fewer candidate sites will receive sufficient sunlight. Landing sites MMO01 and MMO02 stand out because of their favorable Sun and Earth visibility every midsummer until 2032. Sites MMO07 and MMO08 offer access to PSRs in their vicinity that remain below 110 K year-round and should therefore be considered potential destinations for investigating and sampling lunar volatiles contained in icy regolith.

Mons Mouton Plateau (MMP) is the largest of the Artemis CLRs outlined by NASA (2024) for the Artemis III mission (Figure 1). MMP encompasses most of the ∼8 km high Mons Mouton, a nearly flat-topped SPA massif (e.g., Bernhardt et al., 2025) that formed when structurally separated crustal blocks were uplifted to form the SPA basin's rim (Wueller et al., 2025). About 12.7% or 564.8 km^2 of the MMP meet the HLS requirements for landing, which is more suitable landing area than of all other CLRs combined (Table 1). We detailed 11 representative CLSs that are distributed across the plateau. Landing sites MMP02, MMP05, and MMP06 provide access to PSRs with temperatures remaining below 110 K, while MMP03 and MMP08 are within 2 km of PSRs that stay below 125 K year-round (Table 3), which may be ideal for studying the seasonal variation and temperature dependency of volatile accumulation. Our illumination modeling shows that in midsummers 2025, 2028, 2029, 2030, and 2032, most of the 11 CLSs remain sunlit for more than 10 consecutive days, enabling extended surface exploration. However, during midsummers 2027 and 2031, none of the representative CLSs meet the illumination criteria. Nevertheless, MMP's large extent suggests that relocating the CLSs within the HLS-compliant area could still provide viable landing options in those years.

Nobile Rim 1 (NR1) overlaps with the other Artemis CLRs, MMO and MMP, and is located on Mons Mouton, near the western rim of the Nobile crater (Figure 1; NASA, 2022a, 2024). Nine CLSs have been identified along the Nobile crater rim. Candidate sites NR101 and NR103 offer consistently favorable illumination with more than 17 consecutive sunlit days during every midsummer until 2032. Most of the other sites will be accessible during the midsummers of 2025, from 2028 to 2030, and in 2032. Although maximum summer temperatures remain

above 125 K in PSRs (at a Diviner resolution of 240 m/pixel; Williams et al., 2019), subsurface volatiles may still be present under insulating layers of ejecta and regolith.

Nobile Rim 2 (NR2) is located at the northern rim of the Nobile crater where it intersects with an unnamed, ∼55 km diameter crater (Figure 1; NASA, 2022a, 2024). We selected nine potential landing sites that represent the size and distribution of the area that meets the HLS requirements for landing in the NR2 region. Our illumination modeling indicates that none of the landing sites will be suitable for landing during midsummer 2027 and 2031. However, during the midsummers of 2025, 2029, 2030, and 2032, all of the CLSs will provide favorable illumination. Two km^2-sized PSRs near NR204 may maintain surface temperatures below 125 K during the summer season, suggesting potential volatile retention that could be targeted by future surface exploration missions.

The Peak Mear Cabeus B (PCB) is situated between the Cabeus and Drygalski craters on a ∼6 km high massif that is located about 35 km southwest of the Cabeus B crater (Figure 1; NASA, 2024). Among all 16 CLRs, PCB has the smallest area fraction that meets the HLS slope requirement, with only 0.5% or 2.1 km^2 of the total area suitable for landing (Table 1). We identified five CLSs (Figure 2b). However, none of these sites offer access to nearby PSRs that maintain surface temperatures below 125 K throughout the year (Table 3; Williams et al., 2019), limiting the potential to sample and analyze volatiles in the region. Our illumination analyses show that in the midsummers of 2025, 2027, 2031, and 2032, most CLSs will be sufficiently lit to support a 6.5-day surface mission. In contrast, the region is entirely unavailable in midsummer 2026 due to limited sunlight, and in 2028, 2029, and 2030, only one of the five CLSs meets the illumination requirements for landing and traversing.

Peak Near Shackleton (PNS) is located between Shackleton and Slater craters (Figure 1; NASA, 2022a) in an area likely altered by the Shackleton impact ∼3.4 Ga ago (Kring et al., 2021). We detailed six CLSs, including previously identified (NASA, 2020a) and characterized (Kumari et al., 2022) site 007, which has favorable illumination for landing in all midsummers through 2032. Although illumination varies across sites and years, at least three sites remain suitable for landing each year. Landing sites PNS01 and PNS02 offer access to PSRs with extremely low midsummer temperatures below 90 K, whereas sites PNS02 and PNS03 host some of the oldest PSR-hosting craters in this study with model ages of 3.2 and 3.75 Ga, respectively. Volatile estimates suggest substantial abundance in PSRs near the landing sites, with up to ∼645,000 kg at PNS03. In particular, site PNS02 stands out as it hosts an extremely low temperature PSR as well as one of the oldest PSR-hosting craters (see Section 6.3). This results in the highest potential to retain volatiles due to cold temperatures and volatile accumulation over extended geologic timescales. Therefore, the PSRs near site PNS02, which have a cumulative volatile abundance of ∼150,000 kg, should be considered potential targets for future missions aiming to study various volatile compounds and sample a billion years record of volatile delivery and retention on the Moon.

Slater Plain (SP) is located ∼20 km northeast of Slater crater on the lunar farside (Figure 1; NASA, 2024). Among the 16 CLRs, SP stands out with the highest proportion of terrain (19.3% or 77 km^2; Table 1) that meets the HLS slope requirement. We identified 11 CLSs that represent the spatial distribution of the available landing area across the region (Figure 2a). CLS SP11 is within the suggested landing area of Samaddar et al. (2025) that is devoid of any large boulders in Mini-RF radar data. Four of our sites (SP01, SP02, SP10, and SP11) provide access to PSRs within 2 km that remain below 110 K, offering promising targets for investigating potentially volatile-rich regolith. Two additional CLSs (SP04 and SP08; Table 3) have nearby PSRs that can maintain temperatures below 125 K, making them ideal for studying volatile retention. Our illumination modeling shows that during the midsummers of 2025, 2026, 2027, and 2030, most CLSs will receive extended sunlight for over 10 consecutive days. However, during the midsummers of 2028, 2029, 2031, and 2032, only one or two CLSs will meet this threshold, which limits surface mission opportunities during those years.

8. Conclusions

Sampling and analyzing volatiles in polar PSRs were recommended (e.g., National Research Council, 2007) to advance our understanding of their origin, distribution, and stability. Such missions help identify potential volatile reservoirs that could be utilized for a sustained human presence on the Moon (NASA, 2020a). Because little is

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known about polar volatiles (e.g., Kleinhenz et al., 2020), a key science objective of NASA's Artemis is program to explore volatiles and their environment within south polar PSRs (e.g., Artemis III Science Definition Team, 2020; NASA, 2022b).

Therefore, we analyzed the surface slopes of the 16 Artemis CLRs to identify suitable landing areas that meet the HLS's requirements, such as having shallow surface slopes of less than 8° (ideally <5°) and being more than 100 m away from steeper slopes (NASA, 2019). Within those suitable landing areas, we identified 130 representative candidate landing sites (CLSs; Figure 2; Table A1) that all have accessible PSRs within 2 km, since this is the maximum distance an astronaut can explore during a walking extravehicular activity (NASA, 2019).

To ensure that the CLSs receive sufficient sunlight for the 6.5-day surface mission (NASA, 2019), we used an illumination model deriving the maximum number of consecutive illuminated days during the midsummer periods from 2025 to 2032. Considering a buffer before and after the surface operation, we define favorable illumination when the CLS is sunlit for at least 10 consecutive days. Our modeling shows that illumination conditions vary significantly on a temporal and spatial scale among the Artemis CLRs. Thus, depending on the launch year, different CLSs will be available for landing and surface mobility (Table A2). Additionally, our illumination model suggests that there will be only minor limitations in availability when simultaneous Sun and Earth visibility is required to enable direct-to-Earth communication (Figure 5). Consequently, we recommend selecting and ranking multiple landing sites according to their availability for a planned surface mission. This provides flexibility in launch times and ensures that backup sites are available for later mission windows. Some CLSs (e.g., CR06, CR07, and 001) have ≥16 consecutive sunlit days in all studied years and could be considered for longer surface missions.

We identified CLSs that may provide access to the coldest and oldest PSRs by integrating Diviner temperature data (Williams et al., 2019) with a newly developed topographic crater diffusion model (Text S1 in Supporting Information S1; solely applied for the Artemis CLRs CR, CR Extension, dGR1, and PNS). We inferred the PSR ages from the model ages of the PSR-hosting craters, indicating that the majority of PSRs within 2 km of the CLSs are relatively young (85% of the studied craters are younger than 1,500 Ma). To estimate the potential amount of volatiles in PSRs, we examined the delivery of volatiles (i.e., water) by solar wind, micrometeoroids, comets, and asteroids, which are the dominant sources of water over the past ∼1.5 Ga. After volatiles are trapped in PSRs, they can be released and lost through plasma sputtering induced by solar wind, impact vaporization by micrometeoroids, sublimation, and ballistic sedimentation. Although quantification of those processes is still unresolved, we used a range of literature values to identify PSRs with relatively higher potential to harbor volatiles. Trapping and retention of volatiles is largely controlled by surface temperature and PSR age; colder (e.g., FRA04, FRA06, PNS01, PNS02, SP01, and SP10; Table 3) and/or older (e.g., CR05, dGR103, 007, PNS02, and PNS03; Figure 8) PSRs have the greatest preservation potential. Cumulative estimated volatile abundances suggest that some landing sites may provide access to hundreds of tons of water (Figure 9), such as PNS03 (644,952 kg), CR05 (425,582 kg), and PNS02 (150,540 kg), which could sustain NASA's ISRU target for multiple years (Sanders et al., 2008; Sanders & Kleinhenz, 2022). However, compared to the volume of an Olympic swimming pool, these volatile estimates could fill it by only ∼26%, ∼17%, or ∼6%, respectively (Figure 10a).

Our results can be used to select south polar landing sites where astronauts can safely explore and access PSRs that may contain volatiles that can be sampled to illuminate our understanding of their origin and distribution. These new findings will help prepare future missions that could utilize the Moon's resources for a longer-term human presence.

Appendix A

The data in Appendix Table A1 contains the geographic locations of all 130 candidate landing sites (CLSs) that were identified and studied. Table A2 presents the maximum number of consecutive days with Sun or combined Sun and Earth illumination for the 130 CLSs for midsummers from 2025 to 2032, as derived from our illumination modeling.

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Table A1
List of Representative Candidate Landing Sites (CLSs) for the Artemis Candidate Landing Regions (CLRs)

Landing site	Latitude	Longitude
Amundsen Rim (AR)		
AR01	−84.4670	66.9700
AR02	−84.4380	67.9720
AR03	−84.4130	69.2910
AR04	−84.3590	68.8510
AR05	−84.2290	69.8770
AR06	−84.1970	69.1610
AR07	−84.1040	69.6890
AR08	−84.0200	69.9100
AR09	−84.0720	67.8040
AR10	−84.1920	67.2810
Connecting Ridge (CR)		
CR01	−89.5270	−154.5920
CR02	−89.5360	−149.9370
CR03	−89.4140	−114.0950
CR04	−89.3320	−118.5060
CR05	−89.4210	−149.4720
CR06	−89.4440	−137.2910
CR07	−89.4910	−138.3520
CR08	−89.6320	−125.1400
001	−89.4660	−136.9490
Connecting Ridge Extension (CRE)		
CRE01	−89.0110	−94.6100
CRE02	−89.2960	−114.0580
CRE03	−88.7880	−107.3040
CRE04	−88.6740	−109.1940
CRE05	−88.7300	−108.6260
CRE06	−89.1170	−102.6120
de Gerlache Kocher Massif (dGKM)		
dGKM01	−85.5200	−115.3830
dGKM02	−85.7770	−116.5020
dGKM03	−85.8570	−112.6860
dGKM04	−85.8650	−114.4430
dGKM05	−85.7070	−113.8610
dGKM06	−85.4630	−117.5940
dGKM_K2023	−85.7260	−116.8130
de Gerlache Rim 2 (dGR2)		
dGR101	−88.5130	−68.0150
dGR102	−88.9490	−74.2120
dGR103	−88.4450	−62.4910
dGR104	−88.4290	−64.8770
dGR105	−88.6210	−57.1840
011	−88.6740	−68.2280

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Table A1
Continued

Landing site	Latitude	Longitude
de Gerlache Rim 2 (dGR2)		
dGR201	−88.3890	−68.8650
dGR202	−88.0370	−75.0440
dGR203	−88.1410	−71.6960
dGR204	−88.1380	−74.3330
dGR205	−88.0650	−62.0740
dGR206	−87.9470	−62.0980
dGR2_K2023	−88.4500	−67.8590
Faustini Rim A (FRA)		
FRA02	−87.7490	95.0840
FRA03	−87.9630	94.3710
FRA04	−88.0750	93.5490
FRA05	−88.0980	89.7820
FRA06	−88.0580	86.0950
FRA07	−87.9120	82.0720
FRA08	−87.9020	88.0520
Haworth (HW)		
HW01	−86.7680	−20.5140
HW02	−86.8080	−21.7190
HW03	−86.8600	−19.8930
HW04	−86.5680	−21.4730
HW05	−86.6620	−30.5480
HW06	−87.0500	−24.2400
HW07	−86.9320	−23.0200
HW08	−86.9240	−25.8090
HW09	−86.6230	−26.0750
HW10	−86.5270	−14.8760
HW_K2023	−86.9900	−24.2540
Mons Malapert (MMA)		
MMA01	−85.9420	−0.9440
MMA02	−86.0150	3.3820
MMA03	−85.9580	−3.2780
MMA04	−86.3100	−0.4590
MMA_K2023	−85.9920	2.0560
Mons Mouton (MMO)		
MMO01	−85.4160	31.8510
MMO02	−85.4810	29.5490
MMO03	−85.4090	30.4060
MMO04	−85.4090	28.1500
MMO05	−85.3440	28.5940
MMO06	−85.2610	30.1860
MMO07	−85.1950	32.9290

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Table A1
Continued

Landing site	Latitude	Longitude
MMO08	−85.5130	35.7040
MMO09	−85.4660	33.0400
MMO10	−85.4610	31.4900
Mons Mouton Plateau (MMP)		
MMP01	−85.1110	26.7100
MMP02	−84.8050	24.8870
MMP03	−84.9680	31.1530
MMP04	−84.6860	20.8870
MMP05	−84.4160	28.2120
MMP06	−84.5960	30.4520
MMP07	−84.6830	37.3390
MMP08	−84.2110	40.8100
MMP09	−83.6840	32.8850
MMP10	−84.1380	25.6420
MMP11	−84.2690	36.6720
Nobile Rim 1 (NR1)		
NR101	−85.2950	36.9170
NR102	−85.2610	36.8890
NR103	−85.5460	37.5570
NR104	−85.0170	37.8750
NR105	−85.0720	37.5550
NR106	−85.1560	36.9860
NR107	−85.1980	37.1770
NR108	−85.1030	36.7510
NR109	−85.0400	37.4340
Nobile Rim 2 (NR2)		
NR201	−84.0540	56.4810
NR202	−84.1550	59.8350
NR203	−83.6870	57.0080
NR204	−83.7850	57.6830
NR205	−83.6180	59.0570
NR206	−83.7810	59.2820
NR207	−83.9180	59.3260
NR208	−83.9060	57.8110
NR2_K2023	−84.0740	59.1970
Peak Near Cabeus B (PCB)		
PCB01	−83.7630	−69.9830
PCB02	−83.6320	−67.6340
PCB03	−83.5630	−68.2010
PCB04	−83.3690	−67.6650
PCB05	−83.5390	−71.9420

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Table A1
Continued

Landing site	Latitude	Longitude
Peak Near Shackleton (PNS)		
PNS01	−88.9690	134.1060
PNS02	−89.0610	128.4420
PNS03	−88.8520	126.9920
PNS04	−88.9090	120.5380
PNS05	−88.6970	131.0820
007	−88.8110	123.5540
Slater Plain (SP)		
SP01	−87.4780	124.5210
SP02	−87.3820	126.5130
SP03	−87.2830	123.7480
SP04	−87.1920	123.0340
SP05	−87.1820	119.7030
SP06	−87.0730	126.1310
SP07	−87.1310	127.8570
SP08	−87.3270	130.0540
SP09	−87.1840	133.3440
SP10	−87.0320	130.7020
SP11	−86.9500	126.6750

Table A2
Maximum Number of Consecutive Days With Sun or Sun and Earth Illumination for 130 CLS for the Midsummers From 2025 to 2032

	2025		2026		2027		2028		2029		2030		2031		2032	
Landing site	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth
AR01	2	1	**15**	**15**	2	2	8	8	**11**	**10**	2	2	2	1	2	1
AR02	**17**	**17**	**18**	**18**	2	2	8	8	**11**	**10**	**18**	**18**	1	0	**11**	**11**
AR03	**17**	**17**	**19**	**18**	2	2	8	8	**11**	**10**	**19**	**18**	0	0	**11**	**11**
AR04	**15**	**15**	**15**	**15**	2	2	8	8	**11**	**10**	**15**	**15**	0	0	**11**	**11**
AR05	**18**	**17**	**18**	**18**	2	2	9	9	**11**	**10**	**18**	**18**	0	0	**12**	**12**
AR06	2	2	**15**	**15**	3	2	2	2	**11**	**10**	2	2	2	2	2	2
AR07	2	2	**18**	**17**	3	2	3	2	**11**	**10**	**17**	2	3	2	3	2
AR08	**18**	**11**	**19**	**11**	2	2	9	9	**11**	1	**18**	**11**	0	0	**11**	**11**
AR09	1	4	**11**	**11**	4	4	8	8	**10**	**10**	4	3	3	4	4	4
AR10	6	**12**	**12**	**11**	1	1	8	8	**10**	9	**12**	**11**	0	0	**11**	**11**
CR01	3	2	3	3	**17**	3	3	3	4	4	3	3	0	0	3	3
CR02	**19**	**17**	**21**	**20**	4	2	**10**	1	**13**	1	**23**	**21**	2	1	**13**	1
CR03	2	2	5	5	8	8	**15**	**14**	**17**	**16**	3	3	6	6	**17**	**17**
CR04	9	4	8	8	1	1	7	7	8	8	**11**	**11**	**11**	**11**	**11**	**11**
CR05	**12**	**11**	**15**	**15**	**18**	**18**	3	3	6	4	**18**	**18**	0	0	6	6
CR06	**32**	**32**	**21**	**21**	**31**	**31**	**18**	**17**	**18**	**17**	**31**	**31**	**21**	**21**	**30**	**30**
CR07	**32**	**32**	**21**	**21**	**31**	**31**	**18**	**17**	**18**	**17**	**31**	**31**	**21**	**21**	**30**	**30**
CR08	6	6	0	0	7	7	**11**	**11**	**11**	**11**	2	2	5	5	**11**	**11**

Table A2
Continued

Landing site	2025		2026		2027		2028		2029		2030		2031		2032	
	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth
001	**32**	**32**	**21**	**21**	**31**	**31**	**18**	**17**	**18**	**17**	**31**	**31**	**21**	**21**	**30**	**30**
CRE01	**32**	5	**21**	**21**	**30**	**20**	**17**	**17**	**18**	**17**	**31**	**14**	**21**	**18**	**30**	**29**
CRE02	6	5	1	1	**20**	**20**	0	0	1	0	**14**	**14**	**18**	**18**	**22**	**22**
CRE03	6	6	9	9	**18**	**18**	0	0	0	0	**15**	**15**	**18**	**18**	**18**	0
CRE04	6	6	9	9	**19**	**19**	0	0	0	0	**15**	**15**	**19**	**18**	**19**	0
CRE05	3	6	9	9	**10**	9	6	6	6	6	3	2	**10**	9	**10**	1
CRE06	3	2	7	7	2	2	7	7	7	7	8	8	0	0	8	7
dGKM01	3	0	1	0	**21**	**20**	1	0	0	0	**15**	**15**	**18**	**18**	**24**	0
dGKM02	**11**	9	**14**	**13**	**25**	**24**	1	1	3	2	**20**	**19**	**21**	**21**	**20**	**20**
dGKM03	3	1	**12**	6	**16**	**16**	0	0	3	2	**11**	**11**	**14**	**14**	4	3
dGKM04	1	2	3	2	2	2	3	2	3	2	3	2	9	9	2	2
dGKM05	5	9	0	0	9	9	**10**	9	0	0	4	4	8	8	9	9
dGKM06	3	3	3	3	7	7	0	0	0	0	7	6	7	7	7	6
dGKM_K2023	6	1	**13**	**13**	**23**	**21**	3	1	3	2	**23**	**19**	0	0	4	4
dGR101	4	3	7	7	**18**	**18**	**17**	**17**	0	0	**13**	**12**	**16**	**15**	**24**	**19**
dGR102	1	1	3	3	2	2	9	9	**11**	**10**	1	1	**10**	**10**	**11**	**11**
dGR103	**20**	**19**	0	0	**11**	**11**	**17**	**17**	**18**	**17**	5	5	9	9	**20**	**19**
dGR104	**18**	**17**	0	0	9	9	**16**	**15**	**17**	**17**	4	4	7	7	**18**	**17**
dGR105	3	3	7	7	6	6	5	5	6	6	6	6	0	0	6	6
011	**32**	**32**	**21**	**21**	**31**	**31**	**17**	**17**	**18**	**17**	**31**	**31**	**21**	**21**	**30**	**30**
dGR201	6	5	9	9	**20**	**20**	0	0	0	0	**15**	**15**	**18**	**18**	**26**	**26**
dGR202	**18**	**17**	3	3	**15**	**14**	0	0	0	0	9	9	**13**	**12**	**18**	**17**
dGR203	4	3	7	7	**18**	**18**	0	0	0	0	**13**	**13**	**16**	**16**	**23**	**22**
dGR204	4	4	7	7	**19**	**18**	0	0	0	0	**13**	**13**	**17**	**16**	**22**	**22**
dGR205	**16**	**16**	0	0	8	8	8	0	0	0	6	6	**10**	**10**	**16**	**16**
dGR206	2	1	0	0	**11**	0	1	1	**12**	0	3	3	7	7	**10**	**10**
dGR2_K2023	2	5	8	8	**20**	**20**	**17**	**17**	**17**	**17**	**14**	**14**	**18**	**18**	**27**	**27**
FRA01	**17**	**17**	**20**	**20**	2	2	8	8	**11**	**10**	**21**	**21**	0	0	**11**	**11**
FRA02	**20**	**17**	**20**	**20**	2	2	9	9	**11**	**10**	**21**	**21**	0	0	**12**	**12**
FRA03	7	3	**10**	**10**	**10**	**10**	**12**	**11**	**12**	**11**	1	1	**10**	**10**	1	1
FRA04	2	2	8	8	8	7	**11**	**11**	**11**	**11**	1	1	8	7	**11**	**11**
FRA05	6	6	0	0	7	4	**13**	**13**	**13**	2	1	1	5	2	**13**	**13**
FRA06	5	5	**11**	**10**	2	2	9	9	**10**	**10**	**11**	**10**	0	0	**10**	**10**
FRA07	1	1	**12**	**12**	**12**	**12**	0	0	9	8	**11**	**10**	1	1	1	0
FRA08	9	9	0	0	7	7	**14**	**13**	**16**	**15**	2	2	5	5	**16**	**15**
HW01	**16**	**15**	2	2	**25**	1	2	2	4	3	**11**	**10**	**21**	3	5	4
HW02	**23**	**16**	1	1	**13**	**13**	2	2	4	3	7	4	**11**	3	**16**	**15**
HW03	**13**	2	**13**	**13**	**13**	2	2	2	5	3	**13**	**13**	9	3	**13**	2
HW04	7	7	**13**	**13**	3	3	1	1	4	3	**19**	**18**	1	1	4	4
HW05	**16**	**15**	3	3	**14**	**14**	0	0	0	0	9	9	**12**	3	**15**	**15**
HW06	**32**	0	**21**	3	**30**	**15**	**17**	**17**	**18**	**17**	**31**	**10**	**21**	3	**30**	**30**

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Table A2
Continued

Landing site	2025		2026		2027		2028		2029		2030		2031		2032	
	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth
HW07	**32**	**11**	**21**	3	**30**	**15**	**17**	2	**18**	4	**31**	**11**	**21**	3	**30**	0
HW08	3	1	6	3	**15**	**15**	0	0	0	0	**11**	0	1	3	**17**	**17**
HW09	6	6	2	1	**11**	**11**	0	0	0	0	8	7	**11**	3	**11**	**11**
HW10	3	3	1	0	1	0	7	7	1	1	1	0	4	3	0	0
HW_K2023	**32**	**18**	**21**	**3**	**30**	**15**	**17**	**17**	**18**	**17**	**31**	9	**21**	3	**30**	5
MMA01	**14**	**14**	0	0	7	7	**14**	**14**	**14**	**14**	2	2	5	5	**14**	**14**
MMA02	**12**	**11**	0	0	6	5	**12**	**11**	**12**	**11**	**12**	**11**	4	4	**12**	**11**
MMA03	**32**	**32**	**21**	**21**	**30**	**22**	**17**	**17**	**18**	**17**	**31**	**16**	**21**	**20**	**30**	**30**
MMA04	0	0	4	3	**10**	**10**	0	0	0	0	9	9	**10**	**10**	**10**	**10**
MMA_K2023	**32**	**32**	**21**	**21**	**31**	**31**	**17**	**17**	**18**	**17**	**31**	**31**	**21**	**21**	**30**	**30**
MMO01	**32**	**32**	**21**	**21**	**31**	**31**	**17**	**17**	**18**	**17**	**31**	**31**	**21**	**21**	**30**	**30**
MMO02	**32**	**32**	**21**	**21**	**31**	**31**	**17**	**17**	**18**	**17**	**31**	**14**	**21**	**21**	**30**	**29**
MMO03	**15**	**15**	8	0	8	8	**14**	**14**	**16**	**15**	2	2	6	6	**15**	**15**
MMO04	**16**	**16**	0	0	8	8	**15**	**15**	**16**	**15**	2	2	7	6	**16**	**15**
MMO05	**16**	**16**	0	0	9	9	**15**	**15**	**16**	**16**	3	3	7	7	**16**	**16**
MMO06	**15**	**14**	0	0	7	7	**14**	**14**	**15**	**15**	2	2	5	5	**14**	**14**
MMO07	9	9	0	0	4	4	9	9	9	9	9	9	2	2	**10**	9
MMO08	4	0	0	0	2	2	5	5	5	5	9	4	0	0	9	9
MMO09	8	8	**19**	**19**	**22**	**22**	7	7	**10**	9	**23**	**23**	0	0	**10**	**10**
MMO10	9	9	0	0	7	7	**13**	**13**	**16**	**14**	1	1	5	5	**16**	**16**
MMP01	**15**	**14**	0	0	6	6	**13**	**13**	**14**	**14**	1	1	4	4	**14**	**14**
MMP02	**15**	**15**	0	0	6	6	**13**	**12**	**14**	**14**	**15**	**15**	4	4	**15**	**15**
MMP03	8	7	8	8	1	1	8	7	8	8	8	8	0	0	8	7
MMP04	**15**	**15**	0	0	8	7	**14**	**14**	**15**	**14**	2	2	6	5	**15**	**15**
MMP05	**11**	**10**	**11**	**11**	3	3	9	9	**11**	**10**	**11**	**10**	1	1	**10**	**10**
MMP06	**10**	9	**10**	9	3	3	9	9	9	9	**10**	9	1	1	**10**	9
MMP07	**12**	1	2	1	4	4	**11**	0	**12**	1	**12**	1	2	2	**12**	1
MMP08	**14**	4	0	0	5	4	**11**	**11**	**13**	**12**	**14**	**13**	3	2	**14**	4
MMP09	**12**	**12**	0	0	4	4	**11**	**11**	**12**	**12**	**11**	**11**	2	2	**12**	**11**
MMP10	**15**	**15**	0	0	7	6	**13**	**13**	**14**	**14**	1	1	5	4	**15**	**15**
MMP11	**16**	**15**	0	0	6	6	**12**	**12**	**14**	**13**	**16**	**15**	4	4	**15**	**15**
NR101	**32**	**32**	**21**	**21**	**31**	**31**	**17**	**17**	**18**	**17**	**31**	**31**	**21**	**21**	**30**	**30**
NR102	**24**	**23**	0	0	**10**	9	**16**	**16**	**18**	**17**	4	4	8	7	**19**	**19**
NR103	**32**	**32**	**21**	**21**	**31**	**31**	**17**	**17**	**18**	**17**	**31**	**31**	**21**	**21**	**30**	**30**
NR104	**18**	**17**	0	0	6	6	**12**	**12**	**15**	**14**	**16**	**16**	4	4	**15**	**15**
NR105	**19**	**17**	0	0	3	3	**10**	9	**12**	**11**	**18**	**18**	1	1	**13**	**12**
NR106	**19**	2	**16**	**15**	6	3	**12**	2	**14**	2	**19**	2	4	2	**15**	**15**
NR107	**18**	**18**	0	0	3	3	**10**	**10**	**12**	**11**	**18**	**17**	1	1	**13**	**13**
NR108	**17**	**16**	**17**	**16**	2	2	9	9	**11**	**10**	**16**	**16**	0	0	**12**	**11**
NR109	**17**	**16**	0	0	5	5	**12**	**12**	**14**	**13**	**16**	**16**	3	3	**15**	**15**
NR201	**15**	**15**	**14**	**14**	2	2	9	9	**11**	**10**	**14**	**14**	0	0	**12**	**12**

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Table A2
Continued

Landing site	2025		2026		2027		2028		2029		2030		2031		2032	
	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth	Sun	Sun and Earth
NR202	**17**	**17**	**17**	**16**	2	2	9	9	**11**	**10**	**16**	**16**	0	0	**12**	**12**
NR203	**16**	**15**	0	0	3	3	**10**	9	**12**	**11**	**15**	**15**	1	1	**13**	**12**
NR204	1	1	2	1	3	2	9	9	**11**	**10**	3	1	3	2	3	2
NR205	**18**	**18**	0	0	3	3	**20**	**10**	**12**	**11**	**18**	**17**	1	1	**13**	**13**
NR206	**17**	**17**	**17**	0	3	3	9	9	**12**	**11**	**17**	**17**	1	1	**12**	**12**
NR207	**17**	**17**	**17**	**17**	3	3	9	9	**11**	**10**	**17**	**17**	1	1	**12**	**12**
NR208	7	7	**13**	**13**	3	2	9	9	**11**	**10**	**13**	**13**	1	1	**12**	**12**
NR2_K2023	**17**	**16**	**17**	**17**	2	2	9	9	**11**	**10**	**16**	**16**	0	0	**12**	**12**
PCB01	2	2	1	1	**15**	1	0	0	0	0	1	1	**15**	1	1	1
PCB02	**13**	**13**	0	0	8	8	**13**	**13**	**13**	**13**	2	2	6	6	**13**	**12**
PCB03	**17**	**17**	1	1	**13**	**13**	0	0	0	0	8	8	**11**	**11**	**17**	**17**
PCB04	**16**	**16**	3	3	**14**	**14**	0	0	0	0	9	9	**12**	**12**	**16**	**16**
PCB05	2	2	5	5	**15**	**15**	0	0	0	0	**11**	**11**	**15**	**15**	**16**	**15**
PNS01	4	3	7	7	**18**	**12**	0	0	0	0	**13**	**11**	**16**	3	**23**	**11**
PNS02	5	4	8	8	4	3	**11**	9	**11**	**10**	**14**	**11**	2	1	**14**	**12**
PNS03	8	7	**11**	**11**	**22**	**22**	0	0	1	0	**16**	**16**	**20**	**20**	2	2
PNS04	**21**	**21**	0	0	**12**	**11**	**17**	**17**	**18**	**17**	7	6	9	9	**22**	**21**
PNS05	**12**	**12**	**15**	**15**	**18**	**16**	3	3	6	5	**18**	**17**	0	0	6	6
007	**32**	**32**	**21**	**21**	**31**	**31**	**17**	**17**	**18**	**17**	**31**	**31**	**21**	**21**	**30**	**30**
SP01	**15**	**15**	**18**	**18**	**19**	**18**	6	6	8	8	**19**	**18**	0	0	9	9
SP02	3	3	2	2	3	3	**10**	**10**	**11**	**11**	3	3	4	3	3	3
SP03	**11**	**10**	**10**	**10**	**10**	**10**	3	2	4	4	**10**	**10**	0	0	6	6
SP04	**11**	**11**	**12**	**11**	**12**	**11**	5	5	7	7	**12**	**11**	0	0	8	8
SP05	**13**	**13**	**13**	1	**13**	1	5	4	6	6	**12**	1	0	0	8	7
SP06	**18**	**17**	**21**	**18**	**20**	**20**	9	9	**10**	2	**23**	**20**	1	0	**12**	**12**
SP07	**16**	**16**	**19**	**19**	**21**	**21**	7	7	9	9	**22**	**21**	0	0	**10**	**10**
SP08	6	6	9	9	**11**	**11**	0	0	0	0	**11**	**11**	**11**	**11**	1	1
SP09	**13**	**13**	**16**	**16**	**20**	**19**	5	4	6	6	**20**	**19**	0	0	8	7
SP10	**14**	1	**17**	**15**	**17**	**15**	6	4	7	7	**17**	**15**	0	0	9	1
SP11	**15**	**15**	**18**	**18**	**18**	**18**	7	7	8	8	**18**	**17**	0	0	**10**	**10**

Note. For the 6.5-day long Artemis III mission (NASA, 2019), illumination should be provided for at least 10 consecutive days, including a safety buffer (bold numbers).

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The landing site shapefile, the illumination maps for each region and year, the shapefile containing the modeled crater ages, and the PSR shapefile with the respective volatile estimates are available via Zenodo (Wueller et al., 2026): https://doi.org/10.5281/zenodo.17084058. Further details regarding our topographic diffusion model, as well as our estimation of volatile loss by sublimation, can be found in Supporting Information S1. The LOLA RDR data are publicly available at the LOLA PDS node (https://imbrium.mit.edu/). The polar LROC NAC mosaic is available here: https://data.lroc.im-ldi.com/lroc/view_rdr/NAC_POLE_SOUTH (Wagner et al., 2015).

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